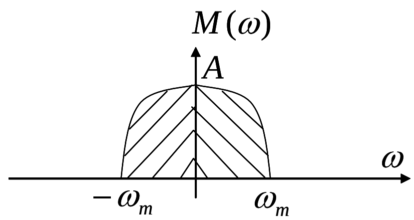
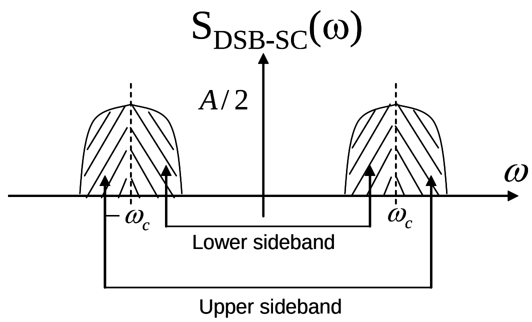
3.6 Single sideband modulation (SSB)

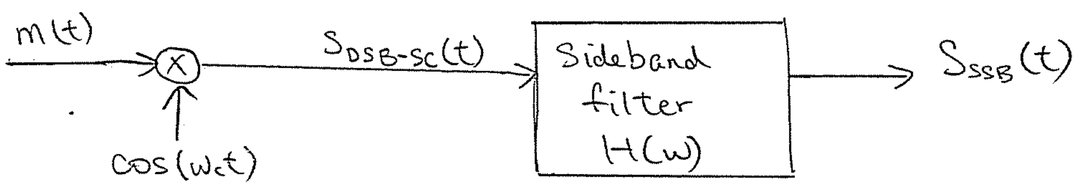
The DSB-SC signal spectrum of a signal *m(t)* has two sidebands: the upper sideband (USB) and the lower sideband (LSB), both containing the complete information of the baseband signal, this is unnecessarily increases the bandwidth. Lower the bandwidth of the modulated signal, more is the number of channels that can be accommodated in a given frequency space. It is, therefore, desirable to transmit only one sideband. This type of modulation is called **single sideband modulation** (SSB).

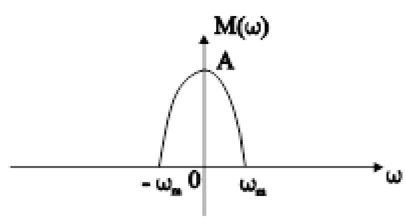


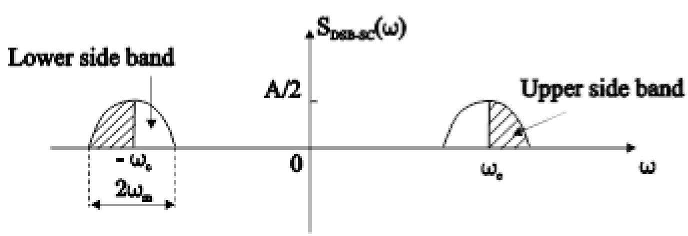


3.6.1. SSB signal generation

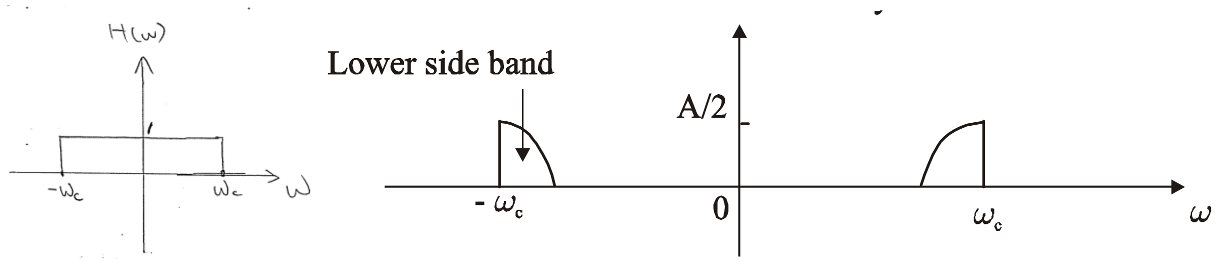
The most straightforward way to generate an SSB signal is to first generate a DSB signal and then suppress one of the sidebands by filtering.



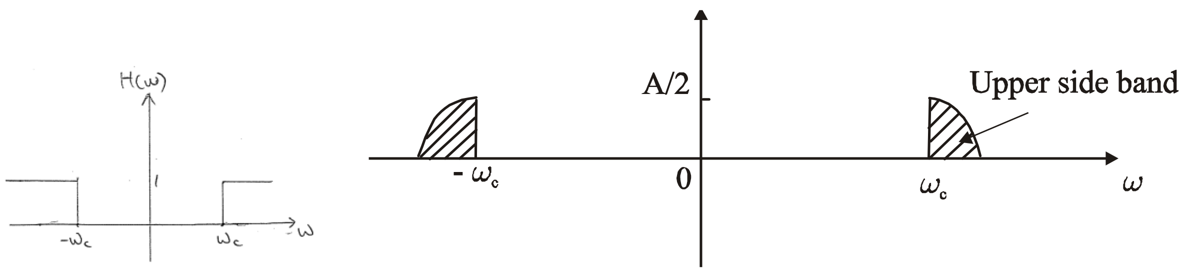




The lower sideband of the signal is obtained by passing $s_{DSB-SC}(t)$ through a low-pass filter of bandwidth ω_c.



To obtain the upper sideband, a high-pass filter with cutoff frequency at ω_c is used.



The primary difficulty of this method is to meet the filter requirement as it has to have a sharp cutoff precisely at the frequency of the carrier wave.

With SSB modulation, the bandwidth of the modulated signal is half of the corresponding DSB-SC signal. Therefore, we can put in twice the number of channels for the same given bandwidth of the channel.

Another method to realize SSB modulation use a procedure that utilizes phase shifting. **Figure 3.11** depicts a system design to retain the lower sidebands. The system $H(\omega)$ in the figure is referred to as a "90° phase-shift network," for which the frequency response is of the form

$$H(\omega) = \begin{cases} -j, & \omega > 0 \\ j, & \omega < 0 \end{cases}.$$

The spectra of $x(t)$, $y_1(t) = x(t)\cos\omega_c t$, $y_2(t) = x_p(t)\sin\omega_c t$, and $y(t)$ are illustrated in **Figure 3.12**. To retain the upper sidebands instead of the lower sidebands, the phase characteristic of $H(\omega)$ is reversed so that

$$H(\omega) = \begin{cases} j, & \omega > 0 \\ -j, & \omega < 0 \end{cases}.$$

Synchronous demodulation of single-side band systems can be accomplished in a manner identical to synchronous demodulation of double-sideband systems. The price paid for the increased efficiency of single-sideband systems is added complexity in the modulator.

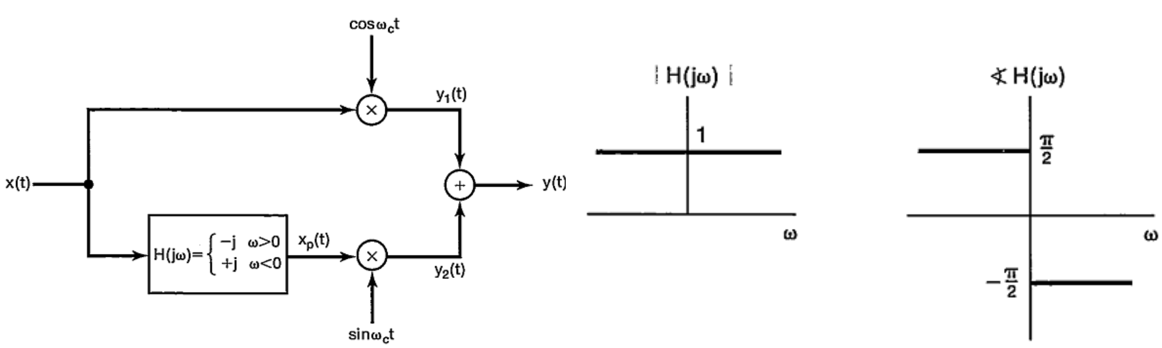


Figure 3.11. System for single-sideband amplitude modulation, using a 90° phase-shift network, in

which only the lower sidebands are retained.

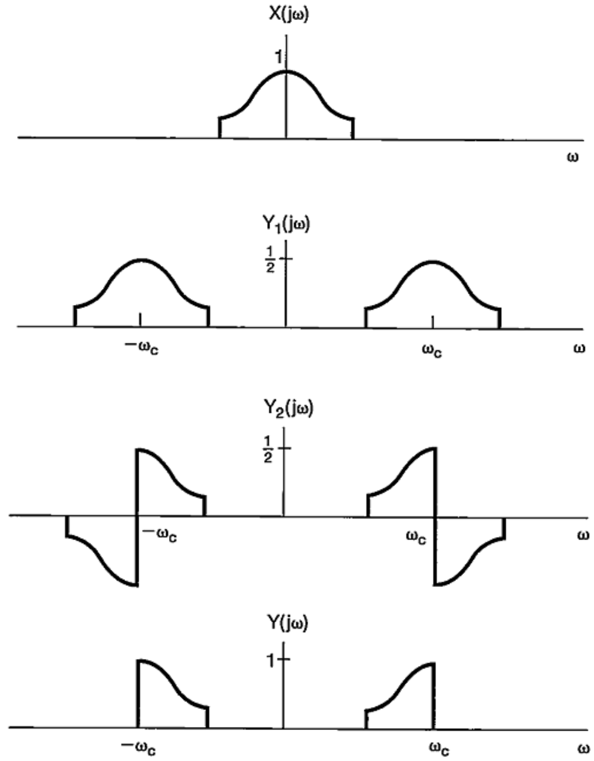


Figure 3.12. Spectra associated with the single-sideband system of **Figure 3.11**.

In summary, throughout this chapter we have seen a number of variations of complex exponential and sinusoidal amplitude modulation. With asynchronous demodulation, a constant must be added to the modulating signal so that it is positive. This results in the presence of the carrier signal as a component in the modulated output, requiring more power for transmission, but resulting in a simpler demodulator than is -required in a synchronous system. Alternatively, only the upper or lower sidebands in the modulated output may be retained, which makes more efficient use of bandwidth and transmitter power, but requires a more sophisticated modulator.

3.7 Quadrature Amplitude Modulation (QAM)

Quadrature amplitude modulation, also referred to as *quadrature amplitude multiplexing*, is a technique in which a signal can be transmitted through a channel without any loss of information. It achieves this by sending 2 DSB signals of the same frequency but offset by 90 degrees ($\pi/2$ radians); this is also known as *Phase Quadrature* (Offsetting by $\pi/2$ radians), hence the name of the modulation technique. **Figure 3.13** shows how this is generally done.

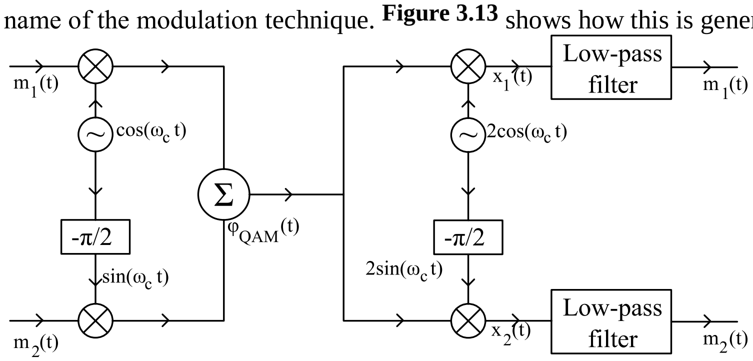


Figure 3.13. Quadrature amplitude modulation.

As shown in **Figure 3.13**, the boxes with the label "-π/2" on them are phase shifters that offset the phase of a signal by -π/2. If a signal of *g(t)cos(t)* is phase shifted by this amount, then it will effectively become *g(t)cos(t - π/2) = g(t)sin(t)*. Therefore, the modulated signal $\varphi_{QAM}(t)$, or alternatively, the sum of the two modulated DSB-signals is effectively:

$$\varphi_{\mathrm{QAM}}(t) = m_1(t) \cos \omega_c t + m_2(t) \sin \omega_c t$$

We can see that both signals have the same frequency, but are now separable because of the phase quadrature. If we consider the multiplier output $x_1(t)$ of the upper arm of the receiver in **Figure 3.13**:

$$\begin{aligned} x_1(t) = 2\varphi_{\mathrm{QAM}}(t) \cos \omega_c t &= 2[m_1(t) \cos \omega_c t + m_2(t) \sin \omega_c t] \cos \omega_c t \\ &= m_1(t) + m_1(t) \cos 2\omega_c t + m_2(t) \sin 2\omega_c t \end{aligned}$$

The last two terms yield a frequency of $2f_c$ (also notice how another QAM signal is formed), and are suppressed by the low pass filter, resulting in the original signal $m_1(t)$. The output of the lower arm of the receiver similarly outputs $m_2(t)$.

$$\begin{aligned} x_2(t) = 2\varphi_{\mathrm{QAM}}(t) \sin \omega_c t &= 2[m_1(t) \cos \omega_c t + m_2(t) \sin \omega_c t] \sin \omega_c t \\ &= m_2(t) - m_2(t) \cos 2\omega_c t + m_1(t) \sin 2\omega_c t \end{aligned}$$

Therefore, two baseband signals of bandwidth *A* Hz, can be utilized by DSB transmission and quadrature multiplexing to transmit it over a bandwidth of 2*A* Hz. The upper channel where $m_1(t)$ is output is called the *in-phase* channel, whereas the lower channel where $m_2(t)$ is output is called the *quadrature* channel, and they are both modulated separately.

A drawback of QAM demodulation is that the process must be almost 100% synchronous. If there is some error in the phase shifting (so if the signal isn't shifted by exactly -π/2 radians), the two channels would interfere with each other and would result in a loss in information. The equation below illustrates this by letting the carrier at the demodulator be $2\cos(\omega_c t + \theta)$:

$$\begin{aligned} x_1(t) &= 2[m_1(t) \cos \omega_c t + m_2(t) \sin \omega_c t] \cos (\omega_c t + \theta) \\ &= m_1(t) \cos \theta - m_2(t) \sin \theta + m_1(t) \cos (2\omega_c t + \theta) + m_2(t) \sin (2\omega_c t + \theta) \end{aligned}$$

Once the low-pass filters suppress the frequencies $2f_c$, the following is the demodulated output:

$$m_1(t) \cos \theta + m_2(t) \sin \theta$$

We can see that we have both input signals (albeit in different proportions) in the demodulated output signal. This interference is undesirable, and similar difficulties come up when other errors are induced such as local frequency error, and unequal attenuation of the upper and lower side band during transmission.

Applications of QAM include the quadrature multiplexing of chrominance signals in analog colour television, WiFi, and digital satellite television transmission.

SSB is similar to QAM with respect to bandwidth requirement, but it is more lenient about the exactness of the carrier frequency or the requirement of a distortion-less transmission medium. SSBs are however, hard to generate if there baseband signal has significant spectral content near the direct current (dc).

3.8 Vestigial Sideband (VSB)

SSB and DSB both have their advantages, and disadvantages. Exact generation of SSB requires a very sharp cut-off, making it difficult. DSB generation is easy to generate, but requires double bandwidth. QAM requires a very precise phase shifter, which can only be approximated.

The *vestigial sideband (VSB) modulation system*, also referred to as *asymmetric sideband*, is a compromise between SSB, and DSB to inherit the advantages of both at a small price. VSB signals are easy to generate, and require just a quarter more bandwidth than SSB.

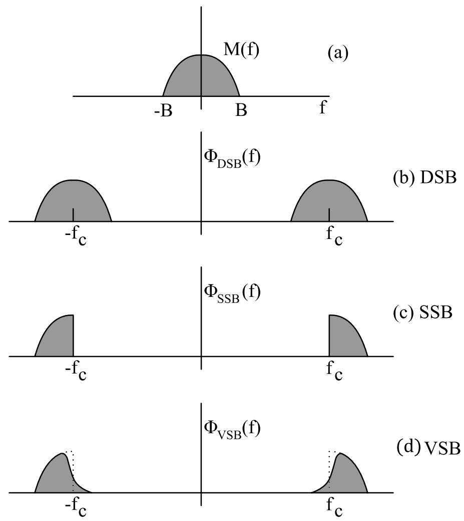


Figure 3.14. Spectra of the modulating signal and corresponding DSB, SSB and VSB signals.

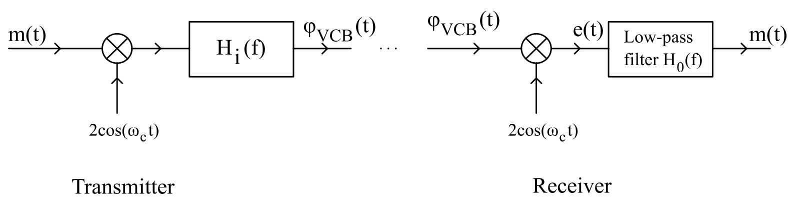


Figure 3.15. VSB modulator and demodulator.

VSB rejects one sideband but only gradually, as shown in **Figure 3.16**. This makes the filter design much easier than that of an SSB filter. The baseband signal can be 100% recovered by using a synchronous detector and equalizer filter $H_0(f)$ at the receiver (**Figure 3.17**).

Generating a VSB out of a DSB uses an arbitrary filter which we will call $H_i(f)$, rendering a VSB signal which looks like the following in the frequency spectrum:

$$\Phi_{\mathrm{VSB}}(f) = [M(f + f_c) + M(f - f_c)]H_i(f)$$

As previously stated, the filter $H_i(f)$ cuts off one side band gradually while retaining the other, making the filter simpler. As a result, VSB bandwidths are a little bit higher than that of SSBs by 25% to 33%.

Since the demodulation of the transmitted signal must yield back the original input, $m(t)$ must be recoverable from $\varphi_{VSB}(t)$ using synchronous demodulation at the receiver. This is achieved by multiplying $\varphi_{VSB}(t)$ by $cos(\omega_c t)$. This will then therefore yield:

$$e(t) = 2\varphi_{\mathrm{VSB}}(t) \cos \omega_c t <==> [\Phi_{\mathrm{VSB}}(f + f_c) + \Phi_{\mathrm{VSB}}(f - f_c)]$$

This signal (*e(t))* is then passed through a low-pass filter $H_0(f)$. Since the output has to be *m(t)*, the spectrum output is thus:

$$M(f) = [\Phi_{\mathrm{VSB}}(f + f_c) + \Phi_{\mathrm{VSB}}(f - f_c)]H_o(f).$$

We can then substitute the expression for $\Phi_{VSB}(t)$ into the above equation. Since the low-pass filter rejects frequencies at an absolute frequency of $2f_c$, we will then obtain:

$$M(f) = M(f)[H_i(f + f_c) + H_i(f + f_c)]\ H_o(f)$$

Following this, we realize that the low-pass filter must satisfy this equation:

$$H_o(f) = \frac{1}{H_i(f + f_c) + H_i\,(f - f_c)}.$$

Note that because $H_i(f)$ is a band-pass filter, the terms $H_i(f \pm f_c)$ contain low-pass components.

Example 3.3
The carrier frequency of a certain VSB signal is f_c = 20 kHz, and the baseband signal bandwidth is 6 kHz. The VSB shaping filter $H_i(f)$ at the input, which cuts off the lower sideband gradually over 2 kHz, is shown in **Figure 3.18**a. Find the output filter $H_o(f)$ required for distortionless reception.

Figure 3.18b shows the low-pass segments of $H_i(f + f_c) + H_i(f - f_c)$· We are interested in this spectrum only over the baseband (the remaining undesired portion is suppressed by the output filter). This spectrum is 0.5 over the band of 0 to 2 kHz, and is lover 2 to 6 kHz, as shown in **Figure 3.18**b. Figure **Figure 3.18**c shows the desired output filter $H_o(f)$, which is the reciprocal of the spectrum in **Figure 3.18**b.

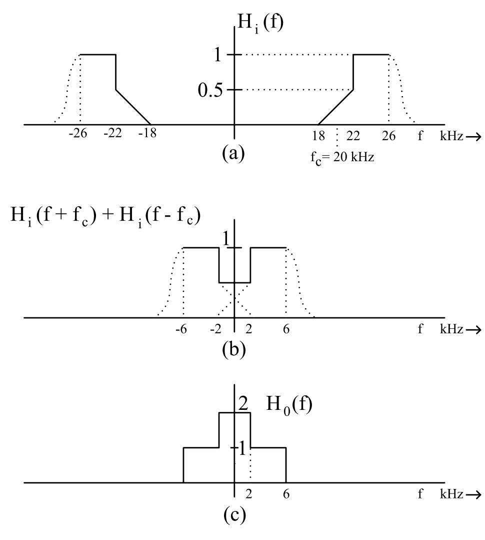


Figure 3.18. VSB modulator and receiver filters.

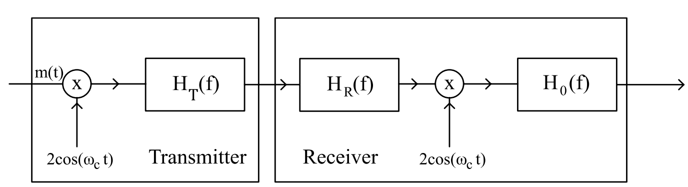


Figure 3.19. Transmitter filter $H_T(f)$, receiver front-end filter $H_R(f)$, and the receiver output low-pass filter $H_o(f)$ in VSB television systems.

3.8.1. Use of VSB in Broadcast Television

VSB is commonly used in television broadcast systems due to the advantages it offers. Baseband video signals for televisions take up a large bandwidth (4.5 MHz), and therefore, DSB would take up 9 MHz. Why not SSB? Well, it turns out that baseband video signals have a good amount of power in low frequency regions, and we know that completely rejecting a sideband is difficult. To reduce receiver costs, it would be desirable if we used an envelope detector over a synchronous detector. SSB+C has low power efficiency, and also SSB would increase receiver cost.

Figure 3.19 shows the process of how Television signals are transmitted and then processed using VSB modulation. As you can see, the filter $H_i(t)$ is made up of 2 other filters, the transmitter RF filter $H_T(f)$, and the receiver RF filter $H_R(f)$. $H_i(t) = H_R(f)H_T(f)$

The requirement of the receiver output filter $H_0(f)$ in a VSB system would therefore still follow.

Figure 3.20 shows how the vestigial shaping filter $H_T(f)$ gradually cuts off the lower sideband of the DSB spectrum gradually from 0.75 MHz to 1.25 MHz below the carrier frequency f_c. This reduces the bandwidth of the VSB signal to 6 MHz, as opposed to DSB and SSB's 9 MHz, and 4.5 MHz each respectively.

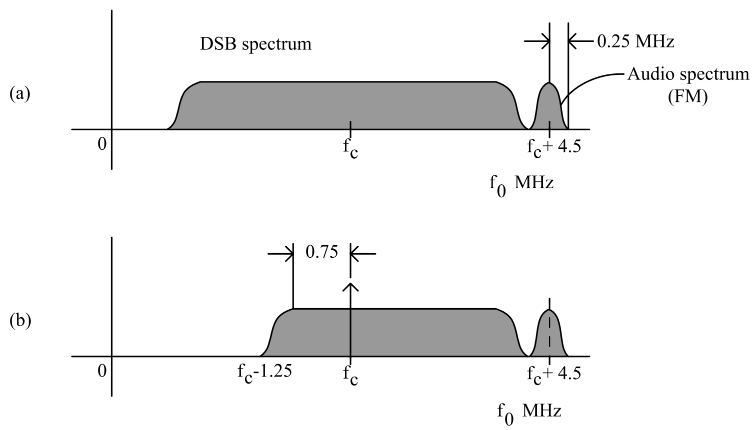


Figure 3.20. Television signal spectra: (a) DSB video signal plus audio; (b) signal transmitted.

3.9 Comparison of various AM systems

Various amplitude systems have been discussed, and it is interesting to compare their performance characteristics.

1. Receiver end

A standard (conventional) AM (large carrier) system has the advantage of simple and cheap envelope detectors. This is why standard AM systems are used more often in public communication system where a single transmitter interacts with many more receivers. SSB-SC, DSB-SC, and QAM receivers are more costly because they require for circuits to synchronize the phase of the carrier at transmitter and receiver.

2. Transmitter end

The SC (suppressed carrier) system need low power transmitters as it transmits sideband power, and not carrier power. Low power transmitters used in SC systems are cheaper than the high power AM transmitters. SC systems are desirable in situations where we need more transmitters and just a few receivers per transmitter such as point to point communication.

3. Generation of modulated signals

SSB-SC signals are more difficult than DSB-SC, and standard AM because the a sharp and precise filter is needed to obtain the SSB signal from a DSB signal. QAM is more complicated than DSB-SC in terms of circuitry, but not by much.

4. Bandwidth

SSB systems are the most advantageous from a bandwidth point of view because the bandwidth of the transmitted signal and the baseband message bandwidth is the same. SSB-SC is used in long-range high-frequency communications, especially in audio communication where the phase-distortion is minimal. DSB-SC and standard AM systems require twice the amount of bandwidth as the original message. QAM can double the messages with the same bandwidth, reducing the bandwidth required per message by a factor of 2. VSB bandwidth requirements is in between SSB-SC, and DSB-SC.

5. Transmission efficiency

DSB, SSB, QAM, and VSB systems are all 100% efficient (meaning that all the transmitted power comes from the message). Standard AM systems however, have a maximum efficiency of tone modulation is 33.3%, corresponding to J = 1.

6. Noise performance

SSB-SC, DSB-SC and QAM systems have the same output signal-to-noise ratio as the baseband system, hence they are equivalent from the noise performance point of view. The standard AM system (with a large carrier) using the envelope detector has maximum $\gamma = 1/3$. (We will cover this later)

In conclusion, we see that each type of amplitude has its advantages and disadvantages. Each circumstance must be evaluated on its own merits and the choice of modulation tailored to that situation.

FREQUENCY TABLE FOR SOUND BROADCASTING SERVICES IN HONG KONG

BROADCASTER	STATION NAME	VHF/FM FREQUENCY (MHz)							MF/AM FREQUENCY (kHz)	
		MOUNT GOUGH	CLOUDY HILL	CASTLE PEAK	GOLDEN HILL	LAMMA ISLAND	BEACON HILL	KOWLOON PEAK	GOLDEN HILL	PENG CHAU
Hong Kong Commercial Broadcasting Co. Ltd. (HKCB)	CR1	88.1	88.3	88.6	88.9	89.1	89.2	89.5	---	---
	CR2	90.3	90.7	91.2	90.9	91.6	91.1	92.1	---	---
	CRE	---	---	---	---	---	---	---	---	864
Radio Television Hong Kong (RTHK)	RTHK1	92.6	93.2	93.4	92.9	93.6	93.5	94.4	---	---
	RTHK2	94.8	95.3	96.4	95.6	96.0	96.3	96.9	---	---
	RTHK3	---	---	---	---	---	---	---	567	---
	RTHK4	97.6	97.8	98.7	98.4	98.2	98.1	98.9	---	---
	RTHK5	---	---	---	---	---	---	---	783	---
	RTHK6	---	---	---	---	---	---	---	---	675
	PUTONGHUA	---	---	---	---	---	---	---	621	---
Metro Broadcast Corporation Ltd. (MB)	Metro Info	99.7	100.0	100.4	101.6	102.1	100.5	101.8	---	---
	Metro Finance	104.0	104.7	102.5	105.5	104.5	102.4	106.3	---	---
	AM1044 Metro Plus	---	---	---	---	---	---	---	---	1044
Effective Radiated Power (kW)		3	0.5	0.7	0.1	0.5	0.15	1	20	10

Note 1 : RTHK3 is also available locally at 97.9 MHz (FM, 20 W transmitter power, Mt. Nicholson), 106.8 MHz (FM, 60 W transmitter power, Chung Hom Kok), 1584 kHz (AM, 100 W transmitter power, Chung Hom Kok) and 107.8 MHz (FM, 15 W transmitter power, Tseung Kwan O; FM, 37.5 W transmitter power, Tin Shui Wai).
RTHK5 is also available locally at 106.8 MHz (FM, 10 W transmitter power, Castle Peak), 99.4 MHz (FM, 15 W transmitter power, Tseung Kwan O), 95.2 MHz (FM, 20 W transmitter power, Mt. Nicholson) and 92.3 MHz (FM, 37.5 W transmitter power, Tin Shui Wai).
RTHK's Putonghua channel is also available locally at 100.9 MHz (FM, 5 W transmitter power, Tai Hang Road; FM, 3 W transmitter power, Castel Peak) and 103.3 MHz (FM, 15 W transmitter power, Tseung Kwan O; FM, 37.5 W transmitter power, Tin Shui Wai).
Metro Info is also available locally at 101 MHz (FM, 10 W transmitter power, Stanley)
Metro Finance is also available locally at 102.6 MHz (FM, 10 W transmitter power, Stanley)

Note 2 : All radio services from HKCB, RTHK and Metro are rebroadcast inside the traffic tunnels as follow.

TRAFFIC TUNNEL	Tai Wai Sha Tin Heights Eagle's Nest Nam Wan	Cross Harbour	Eastern Harbour	Tate's Cairn	Western Harbour	Tai Lam	Lion Rock	Aberdeen	Kai Tak	Shing Mun	Tseung Kwan O	Cheung Tsing	Discovery Bay
REBROADCAST FREQUENCY	FM - Mount Gough AM - Golden Hill & Peng Chau	FM - Mount Gough AM - Golden Hill & Peng Chau	FM - Kowloon Peak AM - Golden Hill & Peng Chau	FM - Kowloon Peak AM - Golden Hill & Peng Chau	FM - Mount Gough AM - Golden Hill & Peng Chau	FM - Golden Hill AM - Golden Hill & Peng Chau	FM - Mount Gough AM - Golden Hill & Peng Chau	FM - Mount Gough AM - Golden Hill & Peng Chau	FM - Mount Gough AM - Golden Hill & Peng Chau	FM - Golden Hill AM - Golden Hill & Peng Chau	FM - Kowloon Peak AM - Golden Hill & Peng Chau	FM - Mount Gough AM - Golden Hill & Peng Chau	FM - Castle Peak AM - Golden Hill & Peng Chau

Table 3.2 Allocation of frequencies for radio stations in Hong Kong (obtained from Office of Telecommunications Authority, Government of HKSAR http:// www.ofta.gov.hk).

Television Broadcasting Network of Hong Kong

(A) Analogue Terrestrial Television

Television Transmitter Location	Television Services / Transmit Frequencies												ERP	Pol	Offset
	Jade			Pearl			Home			World					
	TV CH	Vision Frequency (MHz)	Sound Frequency (MHz)	TV CH	Vision Frequency (MHz)	Sound Frequency (MHz)	TV CH	Vision Frequency (MHz)	Sound Frequency (MHz)	TV CH	Vision Frequency (MHz)	Sound Frequency (MHz)	(W)		
Temple Hill	21	471.25	477.25	25	503.25	509.25	23	487.25	493.25	27	519.25	525.25	10,000	H	0 fl NP
Castle Peak	34	575.25	581.25	38	607.25	613.25	42	639.25	645.25	44	655.25	661.25	1,000	H	+5/3 fl NP
Kowloon Peak	34	575.25	581.25	38	607.25	613.25	42	639.25	645.25	44	655.25	661.25	1,000	H	-5/3 fl NP
Golden Hill	33	567.25	573.25	39	615.25	621.25	43	647.25	653.25	45	663.25	669.25	1,000	H	0 fl NP
Cloudy Hill	49	695.25	701.25	51	711.25	717.25	53	727.25	733.25	55	743.25	749.25	1,000	H	0 fl NP
Lamma Island	49	695.25	701.25	51	711.25	717.25	53	727.25	733.25	55	743.25	749.25	1,500	V	-5/3 fl NP
Stanley	34	575.25	581.25	38	607.25	613.25	42	639.25	645.25	44	655.25	661.25	100	H	0 fl NP
Brick Hill	33	567.25	573.25	39	615.25	621.25	43	647.25	653.25	45	663.25	669.25	100	H	+5/3 fl NP
Pottinger Peak	33	567.25	573.25	39	615.25	621.25	43	647.25	653.25	45	663.25	669.25	100	H	+5/3 fl NP
Chai Wan	48	687.25	693.25	50	703.25	709.25	52	719.25	725.25	54	735.25	741.25	100	H	+5/3 fl NP
Mt. Nicholson	48	687.25	693.25	50	703.25	709.25	52	719.25	725.25	54	735.25	741.25	100	V	-5/3 fl NP
Piper's Hill	49	695.25	701.25	51	711.25	717.25	53	727.25	733.25	55	743.25	749.25	100	H	+5/3 fl NP
Chiu Keng Wan Shan	49	695.25	701.25	51	711.25	717.25	53	727.25	733.25	55	743.25	749.25	0.5	H	0 fl NP
Robin's Nest	48	687.25	693.25	38	607.25	613.25	57	759.25	765.25	54	735.25	741.25	100	V	0 fl NP
Beacon Hill	48	687.25	693.25	50	703.25	709.25	52	719.25	725.25	54	735.25	741.25	2	H	0 fl NP
Hill 275 Lantau	48	687.25	693.25	50	703.25	709.25	52	719.25	725.25	54	735.25	741.25	10	H	0 fl NP
Shek Pik	34	575.25	581.25	38	607.25	613.25	42	639.25	645.25	44	655.25	661.25	1	V	--
Wang Chau	49	695.25	701.25	51	711.25	717.25	53	727.25	733.25	55	743.25	749.25	0.1	H	--
Hong Lok Yuen	33	567.25	573.25	39	615.25	621.25	43	647.25	653.25	45	663.25	669.25	0.1	H	--
Shek Kong	49	695.25	701.25	51	711.25	717.25	53	727.25	733.25	55	743.25	749.25	2	V	-5/3 fl NP
Sheung Yeung Shan	48	687.25	693.25	50	703.25	709.25	52	719.25	725.25	54	735.25	741.25	6	V	+1/12 fl NP
On Lok Road Yuen Long	34	575.25	581.25	38	607.25	613.25	42	639.25	645.25	44	655.25	661.25	0.3	V	--
Hill 297 Yuen Long	48	687.25	693.25	50	703.25	709.25	57	759.25	765.25	54	735.25	741.25	2	H	-5/3 fl NP
Hill 141 Tai Lam Chung	48	687.25	693.25	39	615.25	621.25	52	719.25	725.25	54	735.25	741.25	2	V	+5/3 fl NP
Tai Po Tsai	49	695.25	701.25	51	711.25	717.25	53	727.25	733.25	55	743.25	749.25	0.5	V	-5/3 fl NP
Tsuen Wan	26	511.25	517.25	36	591.25	597.25	41	631.25	637.25	60	783.25	789.25	1	V	0 fl NP
Tai O	33	567.25	573.25	49	695.25	701.25	39	615.25	621.25	55	743.25	749.25	7	V	-5/3 fl NP
Royal Ascot Shatin	33	567.25	573.25	39	615.25	621.25	43	647.25	653.25	45	663.25	669.25	1	V	--
Pokfulam	33	567.25	573.25		--		39	615.25	621.25		--		0.25	V	-5/3 fl NP
Tui Min Hoi Sai Kung	24	495.25	501.25	30	543.25	549.25	26	511.25	517.25	57	759.25	765.25	2	V	0 fl NP
Sham Tseng	41	631.25	637.25	60	783.25	789.25	26	511.25	517.25	57	759.25	765.25	0.5	V	0 fl NP
Lam Tei	32	559.25	565.25		--		58	767.25	773.25		--		0.1	V	-5/3 fl NP
Ying Pun	39	615.25	621.25	45	663.25	669.25	26	511.25	517.25	33	567.25	573.25	4	V	-6/12 fl NP
Tsing Yi	57	759.25	765.25	24	495.25	501.25	60	783.25	789.25	30	543.25	549.25	1	V	-6/12 fl NP
Tseung Kwan O Village	22	479.25	485.25	28	527.25	533.25	30	543.25	549.25	40	623.25	629.25	0.3	V	0 fl NP
Hang Hau Village	26	511.25	517.25	58	767.25	773.25	24	495.25	501.25	56	751.25	757.25	0.3	V	-6/12 fl NP
Tuen Mun	39	615.25	621.25	50	703.25	709.25	45	663.25	669.25	52	719.25	725.25	0.4	V	+6/12 fl NP
Tung Chung	45	663.25	669.25	52	719.25	725.25	26	511.25	517.25	50	703.25	709.25	1	V	+6/12 fl NP

Table 3.3 Allocation of frequencies for Analog TV signals in Hong Kong (obtained from Office of Telecommunications Authority, Government of HKSAR, updated Sept. 19, 2011 http:// www.ofta.gov.hk).

(B) Digital Terrestrial Television

Television Transmitter Location	Television Services / Transmit Frequencies						ERP (W)	Pol	Offset
	11. Home 16. World 81. Jade 84. Pearl		82. J2 Channel 83. I News 85. High Definition Jade Channel		12. Asia 13. TVS 15. CCTV 1 17. SZTV				
	TV CH	Centre Frequency (MHz)	TV CH	Centre Frequency (MHz)	TV CH	Centre Frequency (MHz)			
Temple Hill	22	482.00					1,000	H	
Castle Peak	43*	650.00					320	H	
Kowloon Peak	32	562.00					320	H	
Golden Hill	40*	626.00					320	H	
Cloudy Hill	30	546.00					1,000	H	
Lamma Island	30	546.00					150	V	
Mt. Nicholson	41	634.00	35	586.00	37	602.00	10	V	0 MHz
Piper's Hill	59	778.00					10	H	
Sai Wan Shan (Chai Wan)	28	530.00					10	H	
Brick Hill	26	514.00					10	H	
Sheung Yeung Shan	59	778.00					6	V	
Beacon Hill	58	770.00					0.2	H	
Hill 374 (Yuen Long)	28	530.00					5	V	
Pottinger Peak	56	754.00					10	H	
Robin's Nest	36	594.00					10	V	
Stanley	36	594.00					10	H	
Cheung Chau	58	770.00					2	V	
Hill 141 (Tai Lam Chung)	32	562.00	35	586.00	37	602.00	0.2	V	0 MHz
Tai Po Tsai	40	626.00					0.5	V	
Tai O	52	722.00					0.7	V	

* ERP of TV CH 43 and 40 is 100 W

Note of abbreviation:

ERP	Effective radiated power	Pol	Polarization
FL	Line frequency	TV CH	Television channel number
H	Horizontal polarization	V	Vertical polarization
NP	Non-precision offset		

Table 3.4 Allocation of frequencies for Digital TV signals in Hong Kong (obtained from Office of Telecommunications Authority, Government of HKSAR, updated Sept. 19, 2011 http:// www.ofta.gov.hk).

Chapter 4 Angle and Frequency Modulation

Objectives

- To introduce phase modulation (PM) and frequency modulation (FM)
- To define frequency deviation
- To describe the relationship between PM and FM
- To determine the bandwidth of FM signal
- To explain the methods for generating wideband FM signals
- To explain wideband FM demodulation method
- To compare AM and FM

Reference: Chapter 4 of *Communication Systems* by S. Haykin and M. Moher, 5th edition, John Wiley and Sons.

In the last chapter, we discussed a number of specific amplitude modulation systems in which the modulating signal was used to vary the amplitude of a sinusoidal or a pulse carrier. As we have seen, such systems are amenable to detailed analysis using the frequency-domain techniques we developed in preceding chapters. In another very important class of modulation techniques referred to as frequency modulation (FM), the modulated signal is used to control the frequency of a sinusoidal carrier. Modulation systems of this type have a number of advantages over amplitude modulation systems, as we will show in this chapter.

4.1 Generalized frequency

We would need to talk about a new concept before we continue. Consider a sinusoidal signal with constant peak amplitude A

$$s(t) = A\cos(\omega_c t).$$

The phase of $s(t)$ is given by $\theta(t) = \omega_c t$ and the frequency is ω_c or $d\theta(t)/dt$.

Now, a generalized sinusoidal signal of constant peak amplitude A can be expressed as

$$s(t) = A\cos(\theta(t))$$

where θ(t) is the **instantaneous** or **generalized phase**. In this case, the **instantaneous** or **generalized frequency** is then given by

$$\omega(t) = d\theta(t)/dt. \tag{4.1}$$

We begin our discussion of this chapter by introducing the general notion of *angle modulation.* Angle modulation is the process of varying the total phase angle of a carrier wave in accordance with the instantaneous value of the modulating signal, keeping amplitude of the carrier constant. Angle modulation systems are *not* linear. Angle modulation has the advantage of using peak power capability of a transmitter more efficiently as the amplitude of the carrier does not change. However, angle modulation *requires more complicated transmitters and receivers* as compared to amplitude modulation (AM) systems.

There are two types of angle modulation: Phase modulation (PM) and Frequency modulation (FM).

4.1.1. Phase modulation (PM)

In this type of angle modulation, the phase angle is varied linearly with a modulating signal x(t) about an unmodulated angle $\omega_c t$. This is to say, the instantaneous value of a phase angle θ(t) is equal to the phase of an unmodulated wave $\omega_c t$ plus a time varying component proportional to x(t), i.e.

$$\theta_{PM}(t) = \omega_c t + k_P x(t) \tag{4.2}$$

where k_p is known as *phase sensitivity* of the modulator.

The PM waveform can then be expressed as

$$s_{PM}(t) = A\cos[\omega_c t + k_p x(t)] \tag{4.3}$$

4.1.2. Frequency modulation (FM)

In this type of angle modulation, the instantaneous frequency ω(t) is varied linearly with a modulating signal x(t) about an unmodulated frequency ω_c. In other words, the instantaneous value of the angular frequency ω(t) is equal to the frequency of an unmodulated wave ω_c plus a time varying component proportional to x(t), i.e.

$$\omega_{FM}(t) = \omega_c + k_f x(t) \tag{4.4}$$

where k_f represents the *frequency sensitivity* of the modulator.

The total phase angle of the FM wave can be obtained by integrating the instantaneous frequency ω(t)

$$\theta_{FM}(t) = \int \omega(t)dt = \int [\omega_c + k_F x(t)]dt = \omega_c t + k_F \int x(t)dt$$

The corresponding FM wave is given by

$$s_{FM}(t) = A\cos[\omega_c t + k_F \int x(t)dt] \tag{4.5}$$

4.2 Peak Frequency deviation

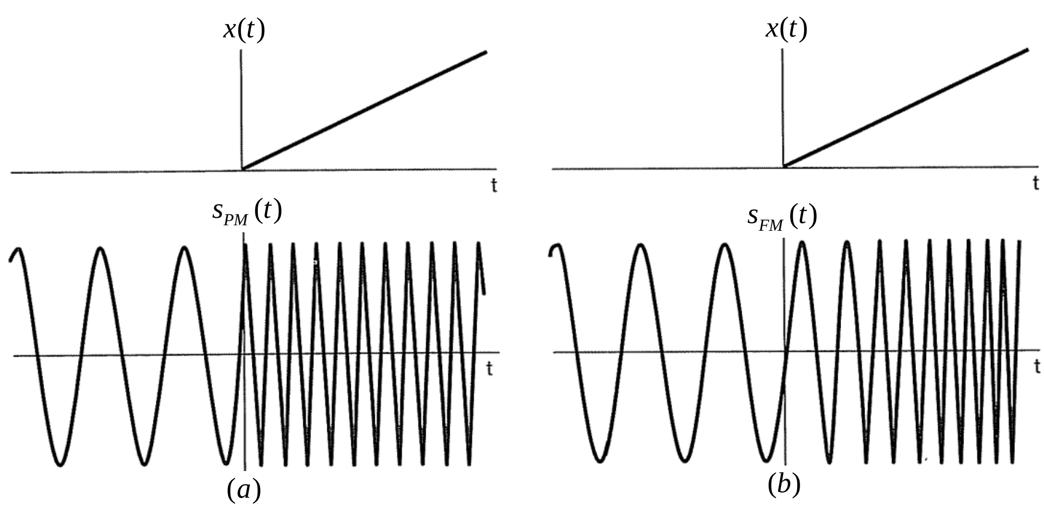
The instantaneous frequency of FM signal, $\omega_{FM}(t) = \omega_c + k_F x(t)$, varies with time. The maximum change in instantaneous frequency from the average, i.e. Δω, is known as **peak frequency deviation**.

$$\Delta\omega = k_f \left| x(t) \right|_{\max}. \tag{4.6}$$

As we will show later, the frequency deviation is a useful parameter for determining the bandwidth of the FM signals.

4.3 Examples of PM/FM modulated waveforms

Illustrations of phase modulated and frequency modulated waveforms are shown in Figure **4.1** (a) and (b). In both cases, the modulating signal is *x(t)* = *tu(t)* (i.e., a ramp signal increasing linearly with time for *t* > 0). In Figure **4.1**(c), an example of frequency modulation is shown with a step (the derivative of a ramp) as the modulating signal [i.e., *x(t)=u(t)*]. Frequency modulation with a step function corresponds to the frequency of the sinusoidal carrier changing instantaneously from one value to another when *x(t)* changes value at *t* =0, much as the frequency of a sinusoidal oscillator changes when the frequency setting is switched instantaneously. When the frequency modulation is a ramp, as in Figure **4.1** (b), the frequency changes linearly with time. Figure 4.2 shows the PM and FM modulated waveform when the message x(t) is a sinusoid. In this case, it is hard to distinguish between PM and FM waveforms.



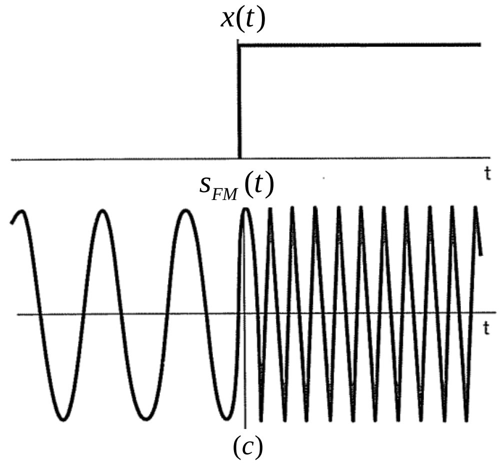


Figure 4.1 Phase modulation, frequency modulation, and their relationship: (a) PM with a ramp as the modulating signal; (b) FM wth a ramp as the modulating signal; (c) frequency modulation with a step (the derivative of a ramp) as the modulating signal.

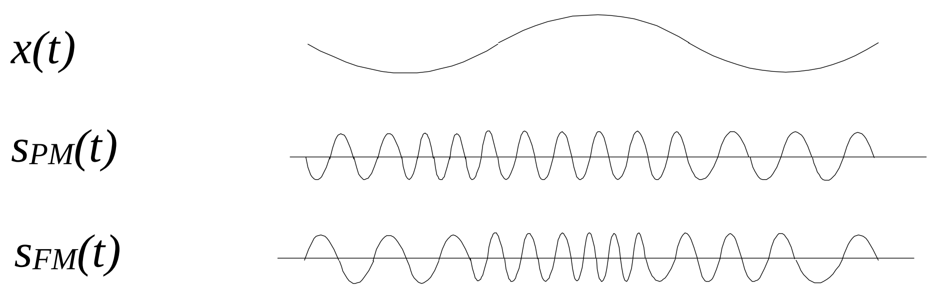
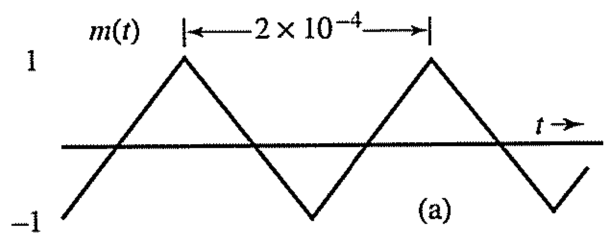


Figure 4.2 PM and FM modulated waveforms when the message signal x(t) is a sinusoid. In this case, *it* is difficult to distinguish PM and FM waveforms.

Example 4.1
Sketch the PM and FM waveform corresponding to the following message signal *m(t).*



4.4 Relationship between PM and FM

PM and FM are actually closely related. From (4.2) and the fact that the instantaneous frequency is the derivative of the phase $\theta(t)$, we can write the instantaneous frequency of the PM waveform as

$$\omega_{PM}(t) = \frac{d\theta_{PM}(t)}{dt} = \omega_c t + k_p \frac{dx(t)}{dt}.$$

Compare with the instantaneous frequency for an FM waveform (Equation (4.3))

$$\omega_{FM}(t) = \omega_c + k_f x(t),$$

we realized that phase modulating the carrier with *x(t)* is identical to frequency modulating with the derivative of *x(t).* Likewise, from (4.5), we see that frequency modulating the carrier with *x(t)* is identical to phase modulating the carrier with the integral of *x(t).* In summary, for PM, the phase angle of the carrier varies linearly with *x(t)* whereas in FM the phase angle varies linearly with the integral of *x(t).* In other words, we can get FM by using PM, provided that at first, the modulating signal is integrated, and then applied to the phase modulator. The converse is also true, i.e. we can generate a PM wave using frequency modulator provided that *x(t)* is first differentiated and then applied to the frequency modulator.

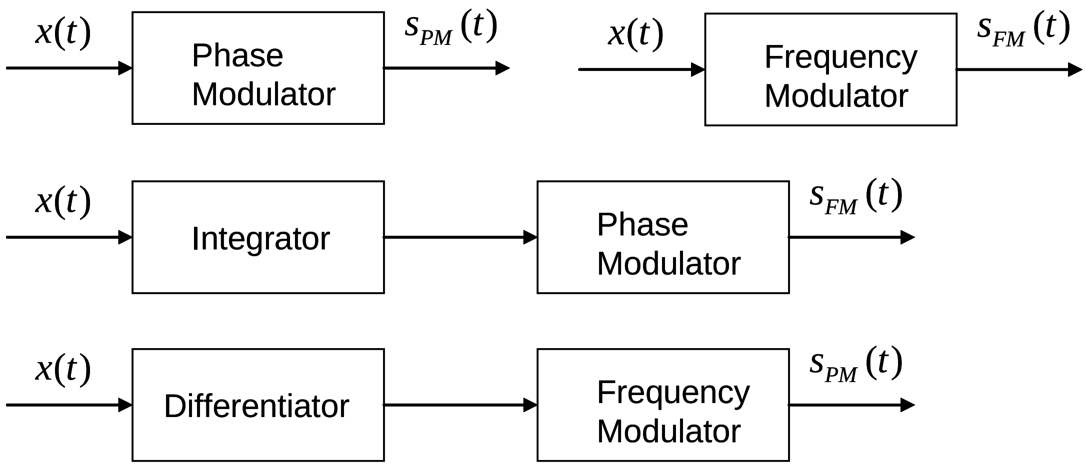


Figure 4.3 Relations between PM and FM waveforms and alternative generation techniques using integrators and differentiators.

4.5 Power of the PM/FM signal

Since the amplitude of PM/FM remains unchanged, the power of PM/FM signal is the same as that of unmodulated carrier.

$$P = \overline{s^2{}_{PM}(t)} = \overline{s^2{}_{FM}(t)} = \overline{A^2 \cos^2\left(\theta(t)\right)} = \frac{A^2}{2}$$

The total power is independent of the PM/FM modulation process since the power is related to signal amplitude, which is constant for PM/FM and not dependent on the signal's phase. In contrast, for AM systems, the peak amplitude of the modulated waveform directly depends on the amplitude of the modulating signal, which can have a large dynamic range-i.e., can vary significantly. Consequently, an PM/FM transmitter can always operate at peak power. In addition, in PM systems, amplitude variations introduced over .a transmission channel due to additive disturbances or fading can, to a large extent, be eliminated at the receiver. For this reason, in public broadcasting and a variety of other contexts, PM/FM reception is typically better than AM reception. On the other hand, we will see that FM generally requires greater bandwidth than AM systems.

4.6 Phase modulation index and Frequency modulation index

Consider the message signal

$$x(t) = A_m \cos\omega_m t$$

is a sinusoid with frequency ω_m. For PM, the modulated waveform is given by

$$s_{PM}(t) = A\cos[\omega_c t + k_P x(t)] = A\cos[\omega_c t + k_P A_m \cos\omega_m t] = A\cos[\omega_c t + m_p \cos\omega_m t]$$

where $m_p = k_P A_m$ is the **phase modulation index**. For FM systems, the output waveform is given by

$$\begin{aligned} s_{FM}(t) &= A\cos[\omega_c t + k_F \int x(t)dt] \\ &= A\cos[\omega_c t + k_F \int A_m \cos\omega_m t dt] \\ &= A\cos[\omega_c t + \frac{k_F A_m}{\omega_m}\sin\omega_m t] \\ &= A\cos[\omega_c t + m_f \sin\omega_m t] \end{aligned}$$

where $m_f = k_F A_m / \omega_m = \Delta\omega / \omega_m$, i.e. the ratio of peak frequency deviation to the modulating frequency, is called the **frequency modulation index**.

FM systems are highly nonlinear and, consequently, are not as straight forward to analyze as are the amplitude modulation systems discussed in previous chapters. However, the methods we have developed in earlier chapters do allow us to gain some understanding of the nature and operation of these systems.

Since FM and PM are easily related, we will phrase the remaining discussion in terms of FM alone. To gain some insight into how the spectrum of the frequency-modulated signal is affected by the modulating signal x(t), it is useful to consider two cases in which the modulating signal is sufficiently simple so that some of the essential properties of frequency modulation become evident.

4.7 Types of frequency modulation

The bandwidth of an FM signal depends on the frequency deviation. When the deviation is high, the bandwidth will be large, and vice-versa. According to the definition of peak frequency derivation $\Delta\omega = k_F|x(t)|_{max}$, for a given message signal *x(t),* the frequency deviation, and hence the bandwidth, will depend on frequency sensitivity k_F. Thus, depending on the value of k_F (or $\Delta\omega$) we can divide FM into two categories: *narrowband FM* and *wideband FM*.

4.7.1. Narrowband FM

When k_F is small, the bandwidth of the FM signal will be narrow. This type of FM is called narrowband FM. Since when $y << 1$, we can approximate $\cos(y) \approx 1$and $\sin(y) \approx y$. When k_F is small, we have

$$\begin{aligned} s_{NBFM}(t) &= A\cos[\omega_c t + k_F \int x(t)dt] \\ &= A\cos(\omega_c t)\cos[k_F \int x(t)dt] - A\sin(\omega_c t)\sin[k_F \int x(t)dt] \\ &\approx A\cos(\omega_c t) - A\sin(\omega_c t) k_F \int x(t)dt \end{aligned} \tag{4.7}$$

Example 4.2

Narrowband FM with tone modulation

Let the message signal be

$$x(t) = A_m \cos\omega_m t.$$

The signal waveform is given by

$$\begin{aligned} s_{NBFM}(t) &= A\cos\omega_c t - A\sin\omega_c t \cdot k_F \int x(t)dt \\ &= A\cos\omega_c t - Ak_F \sin\omega_c t \cdot \int A_m \cos\omega_m t dt \\ &= A\cos\omega_c t - Ak_F \frac{A_m}{\omega_m}\sin\omega_m t \sin\omega_c t \\ &= A\cos\omega_c t - Am_f \sin\omega_m t \sin\omega_c t \\ &= A\cos\omega_c t + \frac{1}{2}Am_f \cos(\omega_c + \omega_m)t - \frac{1}{2}Am_f \cos(\omega_c - \omega_m)t \end{aligned}$$

where $m_f = k_F A_m / \omega_m$ is thefrequency modulation index. The spectrum of $s_{NBFM}(t)$ is then given by

$$\begin{aligned} S_{NBFM}(\omega) &= \pi A\left[\delta(\omega-\omega_c) + \delta(\omega+\omega_c)\right] \\ &\quad + \frac{1}{2}\pi Am_f \left[\delta(\omega-\omega_c-\omega_m) + \delta(\omega+\omega_c+\omega_m) - \delta(\omega-\omega_c+\omega_m) - \delta(\omega+\omega_c-\omega_m)\right]. \end{aligned}$$

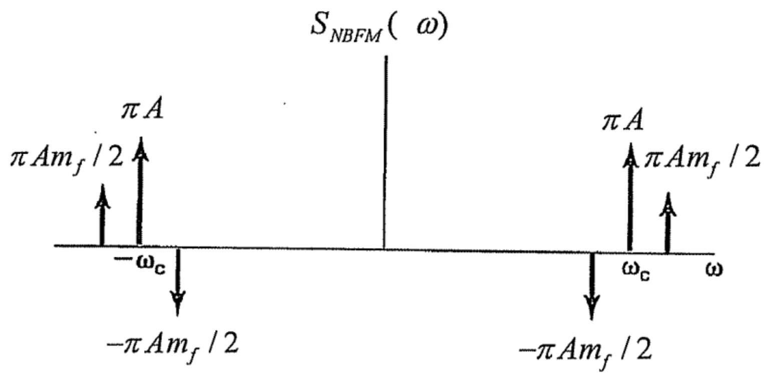


Figure 4.4 Approximate spectrum of a narrowband FM signal with tone modulation.

4.7.2. Wideband FM

When k_F is large, the bandwidth of the FM signal will become wide and this type of FM is called *wideband FM*. It is usually very difficult to analyze a wideband FM signal and therefore we will just briefly talk about wideband FM with tone modulation.

Example 4.3
Wideband FM with tone modulation
Let the message signal be

$$x(t) = A_m \cos\omega_m t.$$

The signal waveform is then given by

$$\begin{aligned} s_{WBFM}(t) &= A\cos[\omega_c t + k_F \int x(t)dt] \\ &= A\cos[\omega_c t + k_F \int A_m \cos\omega_m t dt] \\ &= A\cos[\omega_c t + m_f \sin\omega_m t] \\ &= A\cos\omega_c t\cos(m_f \sin\omega_m t) - A\sin\omega_c t\sin(m_f \sin\omega_m t). \end{aligned}$$

For wideband FM, the approximation $\cos(y) \approx 1$and $\sin(y) \approx y$ will not hold. In this case, since x(t) is periodic with period $T = 2\pi/\omega_m$, $s_{FM}(t)$ will also be periodic with period $2\pi/\omega_m$ and we can express $s_{WBFM}(t)$ as a Fourier Series

$$s_{WBFM}(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_m t}$$

where

$$c_n = \frac{1}{T}\int_{-T/2}^{T/2} s_{WBFM}(t)e^{-jn\omega_m t}dt = \frac{1}{T}\int_{-T/2}^{T/2} A\cos[\omega_c t + m_f \sin\omega_m t]e^{-jn\omega_m t}dt.$$

Now, let us just note that such integral is hard to evaluate and there is no close form solution to this integral (That's why it is much harder to analyze FM systems and AM systems). However, we can still gain some insights from numerically evaluating such integrals. Figure 4.7 shows the magnitude spectrum of $s_{WBFM}(t)$ as a function of the frequency modulation index m_f.

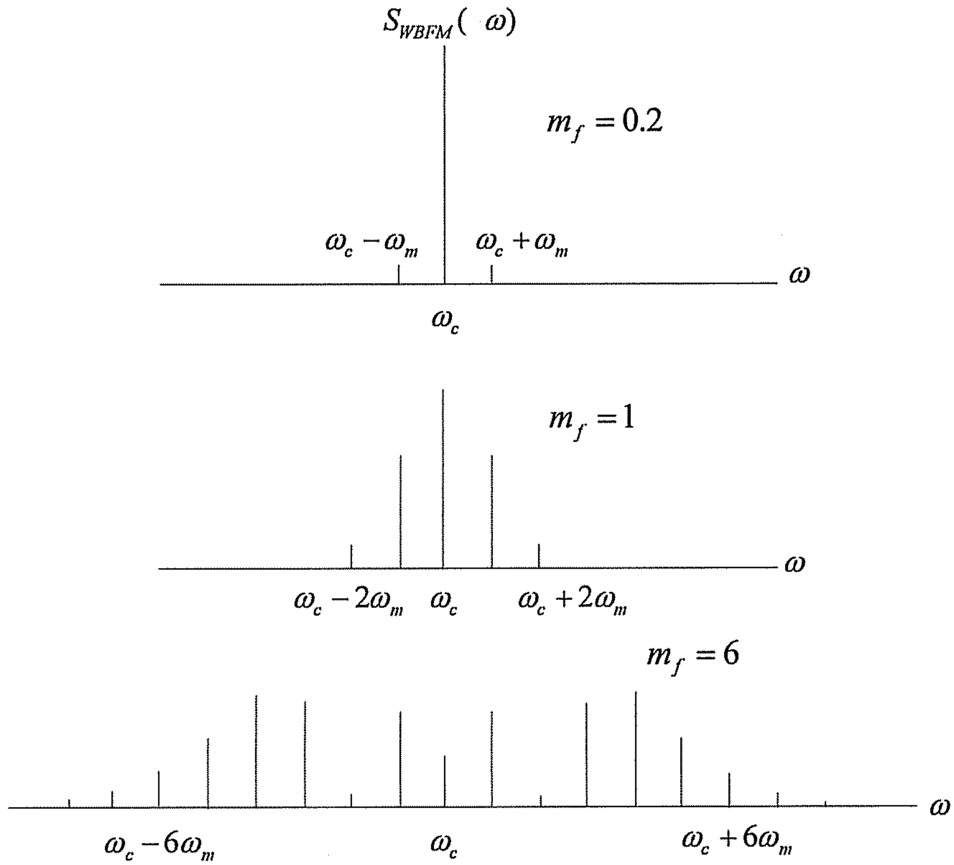


Figure 4.5. Typical plots of $S_{WBFM}(\omega)$ for different m_f.

The following observations can be made

- Theoretically, infinite number of sidebands are produced. The presence of infinite number of sidebands makes the theoretical bandwidth of the FM signal infinite.
- When m_f is small, there are *few sideband frequencies of large amplitude* and, when *m_f is large*, there are *many sideband frequencies but with smaller amplitudes*. Hence, in practice, to determine the bandwidth, it is only necessary to consider a finite number of *significant sideband* components. The sidebands with small amplitudes can be ignored

4.8 Bandwidth of wideband FM signal

As a rule of thumb, the amplitude of the sidebands diminishes rapidly for $n > m_f$, particularly as m_f becomes large, the number of significant sideband is $m_f + 1$, i.e.

$$B_{FM} = 2(m_f + 1)\omega_m = 2(\Delta\omega + \omega_m)$$

Expressed in words, *the bandwidth of an FM signal is twice the sum of the maximum frequency deviation and the bandwidth of the message signal*. This rule for bandwidth is called **Carson's rule**. This formula is not restricted to tone modulation and can be generalized to arbitrary message signals x(t) with bandwidth ω_m.

4.9 FM modulation and demodulation methods

4.9.1. Narrowband FM modulation and demodulation

From the previous section, we discussed that a narrowband FM signal is given by (equation (4.7))

$$s_{NBFM}(t) = A\cos(\omega_c t) - A\sin(\omega_c t)k_F \int x(t)dt\ .$$

The following figures illustrate how narrowband FM signals can be modulated and demodulated.

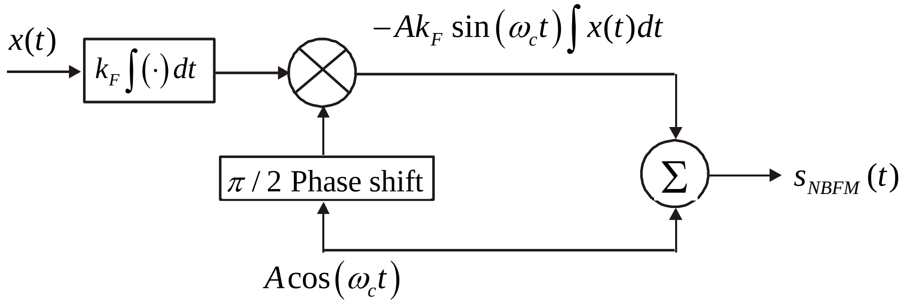


Figure 4.6. Block diagram for the generation of narrowband FM signals.

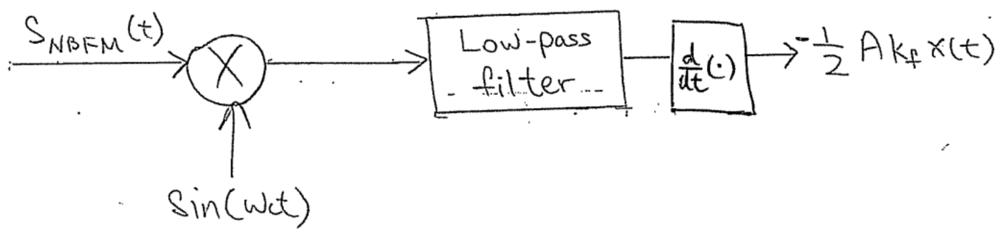


Figure 4.7. Block diagram for the demodulation of narrowband FM signals.

4.9.2. Wideband FM modulation and demodulation

One way to generate a wideband FM signal is to use a oscillator in which the oscillating frequency depends on x(t).

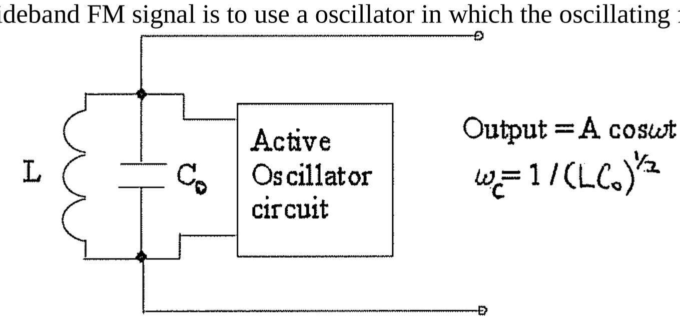


Figure 4.8. Circuit for generating wideband FM signals.

In the voltage-controlled oscillator (VCO), the oscillating frequency is given by

$$\omega_c = \frac{1}{\sqrt{LC_0}}$$

which is also used as the carrier frequency of the FM system. The capacitance of the circuit varies linearly with the message signal x(t) according to

$$C = C_0 + \Delta C = C_0 + k_0 x(t)$$

where k_0 is a constant. The corresponding change in the oscillator frequency will then be given by

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{LC_0(1+\frac{\Delta C}{C_0})}} = \frac{1}{\sqrt{LC_0}}\frac{1}{\sqrt{1+\frac{\Delta C}{C_0}}} \approx \omega_c(1-\frac{\Delta C}{2C_0}) = \omega_c(1-\frac{k_0 x(t)}{2C_0}) = \omega_c - \frac{\omega_c k_0}{2C_0}x(t) = \omega_c - kx(t)$$

where $k = \omega_c k_0/2C_0$ is a constant and the approximation $(1 + y)^{-1/2} = 1 - y/2$ for x small is used.

For the demodulation of wideband FM signals, we can apply $s_{WBFM}(t)$ to a differentiator. The corresponding output will be given by

$$\frac{ds_{WBFM}(t)}{dt} = \frac{d}{dt}\left\{A\cos[\omega_c t + k_F\int x(t)dt]\right\} = -A[\omega_c + k_F x(t)]\sin[\omega_c t + k_F\int x(t)dt]$$

which is similar to a standard AM signal with envelope $-A[\omega_c + k_F x(t)]$. Since $\Delta\omega = k_F|x(t)|_{max} << \omega_c$ usually in practice, we can pass the differentiated output to an envelope detector and the corresponding output (the envelope) will be

$$-A[\omega_c + k_F x(t)]$$

The message signal can then be obtained by eliminating the DC term $A\omega_c$.

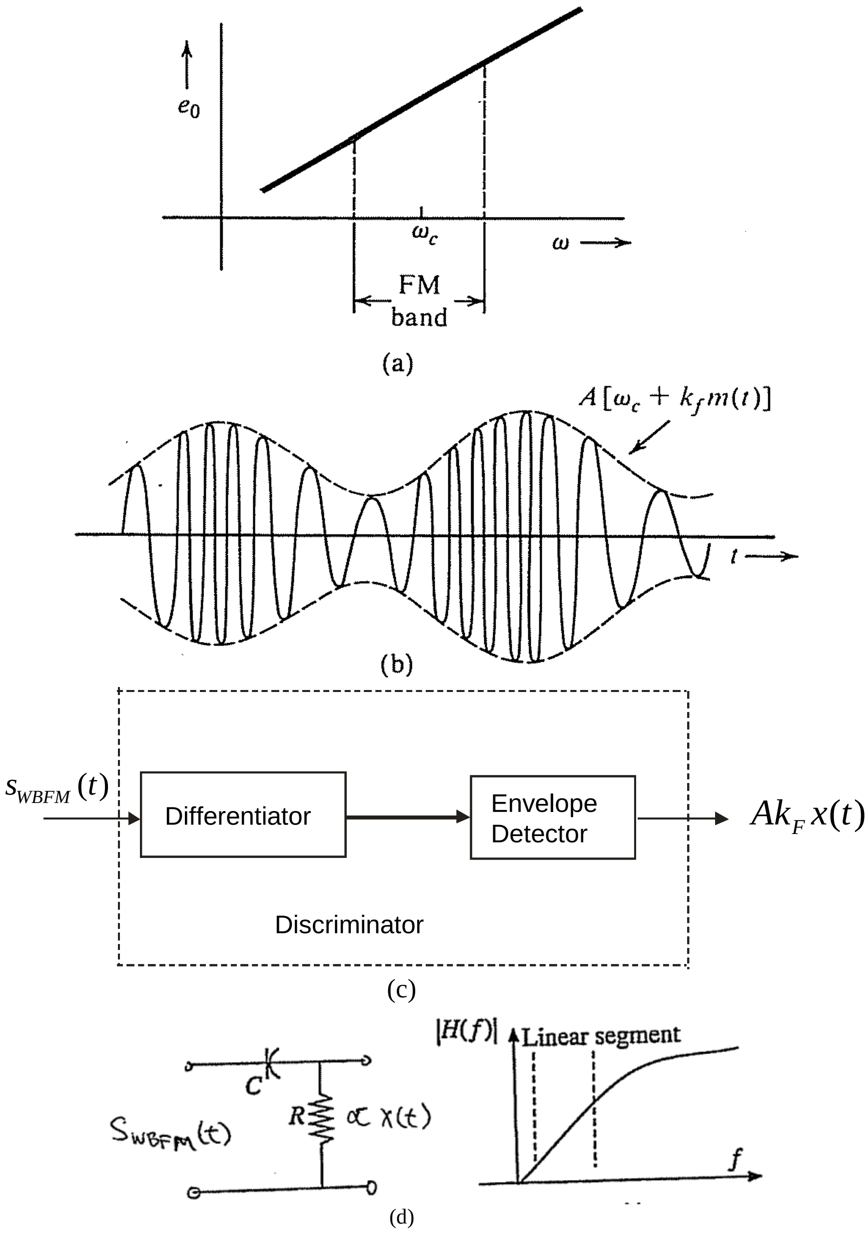


Figure 4.9. a) FM demodulator frequency response. b) Output of a differentiator to the input FM wave. c) Block diagram for the demodulation of wideband FM signals. d) A simple-high pass filter that can be used for FM demodulation.

Circuits that converts the frequency modulated (FM) signal into a corresponding amplitude modulated (AM) signal by using frequency dependent circuits whose output voltage depends on input frequency are called *frequency discriminators*.

4.10 Comparison between AM and FM

Channel bandwidth

The wideband FM has a larger bandwidth as compared to AM because wideband FM produces a larger number of sidebands. In a typical broadcast system, each channel bandwidth in AM is 15kHz, whereas, in FM, it is 150kHz. Therefore, FM has a disadvantage over AM in this aspect.

FREQUENCY TABLE FOR SOUND BROADCASTING SERVICES IN HONG KONG

BROADCASTER	STATION NAME	VHF/FM FREQUENCY (MHz)							MF/AM FREQUENCY (kHz)	
		MOUNT GOUGH	CLOUDY HILL	CASTLE PEAK	GOLDEN HILL	LAMMA ISLAND	BEACON HILL	KOWLOON PEAK	GOLDEN HILL	PENG CHAU
Hong Kong Commercial Broadcasting Co. Ltd. (HKCB)	CR1	88.1	88.3	88.6	88.9	89.1	89.2	89.5	---	---
	CR2	90.3	90.7	91.2	90.9	91.6	91.1	92.1	---	---
	CRE	---	---	---	---	---	---	---	---	864
Radio Television Hong Kong (RTHK)	RTHK1	92.6	93.2	93.4	92.9	93.6	93.5	94.4	---	---
	RTHK2	94.8	95.3	96.4	95.6	96.0	96.3	96.9	---	---
	RTHK3	---	---	---	---	---	---	---	567	---
	RTHK4	97.6	97.8	98.7	98.4	98.2	98.1	98.9	---	---
	RTHK5	---	---	---	---	---	---	---	783	---
	RTHK6	---	---	---	---	---	---	---	---	675
	PUTONGHUA	---	---	---	---	---	---	---	621	---
Metro Broadcast Corporation Ltd. (MB)	Metro Info	99.7	100.0	100.4	101.6	102.1	100.5	101.8	---	---
	Metro Finance	104.0	104.7	102.5	105.5	104.5	102.4	106.3	---	---
	AM1044 Metro Plus	---	---	---	---	---	---	---	---	1044
Effective Radiated Power (kW)		3	0.5	0.7	0.1	0.5	0.15	1	20	10

Note 1 : RTHK3 is also available locally at 97.9 MHz (FM, 20 W transmitter power, Mt. Nicholson), 106.8 MHz (FM, 60 W transmitter power, Chung Hom Kok), 1584 kHz (AM, 100 W transmitter power, Chung Hom Kok) and 107.8 MHz (FM, 15 W transmitter power, Tseung Kwan O; FM, 37.5 W transmitter power, Tin Shui Wai).
RTHK5 is also available locally at 106.8 MHz (FM, 10 W transmitter power, Castle Peak), 99.4 MHz (FM, 15 W transmitter power, Tseung Kwan O), 95.2 MHz (FM, 20 W transmitter power, Mt. Nicholson) and 92.3 MHz (FM, 37.5 W transmitter power, Tin Shui Wai).
RTHK's Putonghua channel is also available locally at 100.9 MHz (FM, 5 W transmitter power, Tai Hang Road; FM, 3 W transmitter power, Castel Peak) and 103.3 MHz (FM, 15 W transmitter power, Tseung Kwan O; FM, 37.5 W transmitter power, Tin Shui Wai).
Metro Info is also available locally at 101 MHz (FM, 10 W transmitter power, Stanley)
Metro Finance is also available locally at 102.6 MHz (FM, 10 W transmitter power, Stanley)

Note 2 : All radio services from HKCB, RTHK and Metro are rebroadcast inside the traffic tunnels as follow.

TRAFFIC TUNNEL	REBROADCAST FREQUENCY
Tai Wai Sha Tin Heights Eagle's Nest Nam Wan	FM - Mount Gough AM - Golden Hill & Peng Chau
Cross Harbour	FM - Mount Gough AM - Golden Hill & Peng Chau
Eastern Harbour	FM - Kowloon Peak AM - Golden Hill & Peng Chau
Tate's Cairn	FM - Kowloon Peak AM - Golden Hill & Peng Chau
Western Harbour	FM - Mount Gough AM - Golden Hill & Peng Chau
Tai Lam	FM - Golden Hill AM - Golden Hill & Peng Chau
Lion Rock	FM - Mount Gough AM - Golden Hill & Peng Chau
Aberdeen	FM - Mount Gough AM - Golden Hill & Peng Chau
Kai Tak	FM - Mount Gough AM - Golden Hill & Peng Chau
Shing Mun	FM - Golden Hill AM - Golden Hill & Peng Chau
Tseung Kwan O	FM - Kowloon Peak AM - Golden Hill & Peng Chau
Cheung Tsing	FM - Mount Gough AM - Golden Hill & Peng Chau
Discovery Bay	FM - Castle Peak AM - Golden Hill & Peng Chau

Table 4.1 Allocation of frequencies for radio stations in Hong Kong (obtained from Office of Communications Authority, Government of HKSAR http://www.ofca.gov.hk/).

Chapter 5 Image signal representation and Fourier Transform

Objectives

- To study basic discrete and continuous 2-dimensional signals
- To explain the concept of spatial signals and spatial frequency
- To explain the representation of 2-D images in terms of 2-dimensional signals
- To describe the properties of spectral amplitude and phase in images
- To describe LTI systems and Fourier Transform in higher-dimensions
- To describe image representation based on Fourier Transform and JPEG format

5.1 Introduction

In this chapter we will study 2-dimensional signals that is used for image representation. We will talk about 2-D LTI systems and Fourier Transform, basic image signal processing and how Fourier Transform is used in JPEG formats.

5.2 Two-dimensional signals and image representation

A 2-D discrete signal $s[n_1, n_2]$ is mathematically a complex *bi-sequence*, or a mapping of the 2-D integers into the complex plane. Our convention is that the signal *s* is defined for all finite values of its integer arguments n_1, n_2 using zero padding as necessary. Occasionally we will deal with finite extent signals, but we will clearly say so in such instances. We will adopt the simplified term *sequence* over the more correct bi-sequence.

5.2.1. Impulse and Delta functions

A simple example of a 2-D signal is the *impulse or Delta function* $\delta[n_1,n_2]$ in 2 dimensions, defined as follows:

$$\delta[n_1,n_2] \triangleq \begin{cases} 1 \text{ for } (\mathrm{n}_1,\mathrm{n}_2) = (0,0) \\ 0 \text{ for } (\mathrm{n}_1,\mathrm{n}_2) \neq (0,0) \end{cases}$$

a portion of which is plotted in Figure 5.1.

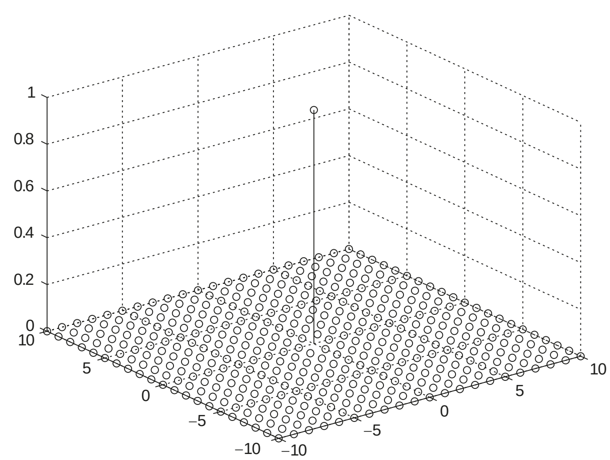


Figure 5.1. 2-D spatial impulse bi-sequence $\delta[n_1,n_2]$.

A general 2-D signal can be written as an infinite sum over shifted impulses and delta functions, very much the same formula we derived back in *Chapter 2.4* i.e.

$$x[n_1,n_2] = \sum_{k_1,k_2=-\infty}^{\infty} x[k_1,k_2]\delta[n_1-k_1,n_2-k_2]$$

where the summation is taken over all integer pairs (k_1, k_2). By the definition of the 2-D impulse, for each point (n_1, n_2), there is exactly one term on the right side that is nonzero, the term $x[n_1, n_2]$, and hence the result (1.1-1) is correct. This equality is called the *shifting* representation of the signal *x*.

Another basic 2-D signal is the impulse line, e.g., a straight line at $45°$,

$$\delta[n_1 - n_2] = \begin{cases} 1, & n_1 = n_2, \\ 0, & \text{else.} \end{cases}$$

A portion of this line is sketched in Figure 5.2.

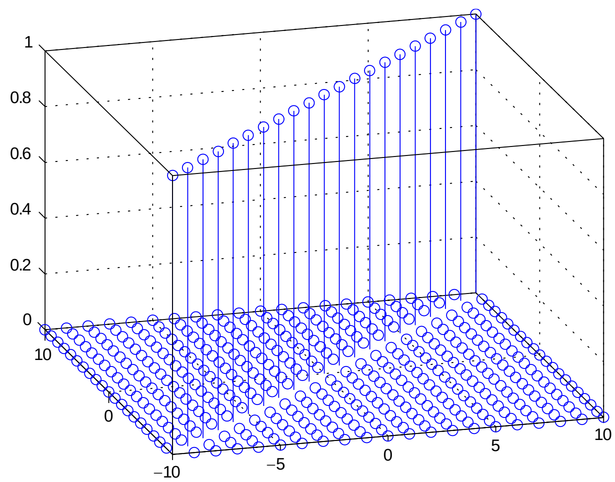


Figure 5.2. A portion of the unit impulse line $\delta[n_1 - n_2]$ at 45°

5.2.2. Step function

Another basic 2-D signal class is step functions, perhaps the most common of which is the first-quadrant *unit step function* $u[n_1, n_2]$, defined as

$$u[n_1, n_2] \triangleq \begin{cases} 1, & n_1 \geq 0, n_2 \geq 0, \\ 0, & \text{elsewhere.} \end{cases}$$

A portion of this function is plotted in Figure 5.3.

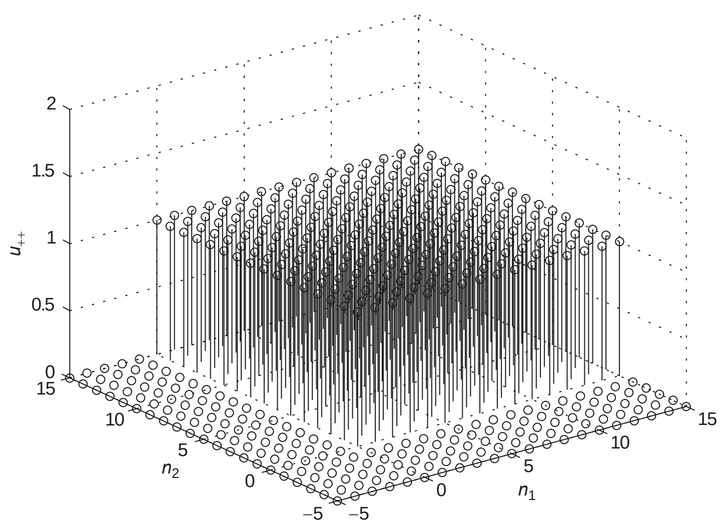


Figure 5.3. A portion of the 2-D unit step function $u[n_1, n_2]$.

5.2.3. Periodic signals

A 2-D signal $x[n_1, n_2]$ is *periodic* with period ($N1$, $N2$) (also written as $N1 \times N2$), if

$$x[n_1, n_2] = x[n_1 + N_1, n_2], \\ = x[n_1, n_2 + N_2]$$

for all integers $n_1$and n_2 where $N1$ and $N2$ are positive integers.

The period ($N1$,$N2$) defines a 2-D grid (either spatial or space-time) over which the signal repeats or is periodic. Since we are in discrete space, the period must be composed of *positive integers*.

Example 5.1

Consider the sine wave

$$x[n_1,n_2] = \sin(2\pi n_1/4).$$

The period in the first dimension is $N1 = 4$. In the vertical direction, the signal is constant though. So we can use any positive integer $N2$, and we choose the smallest such value, $N2 = 1$. Thus the rectangular period is $N1 \times N2 = 4 \times 1$,

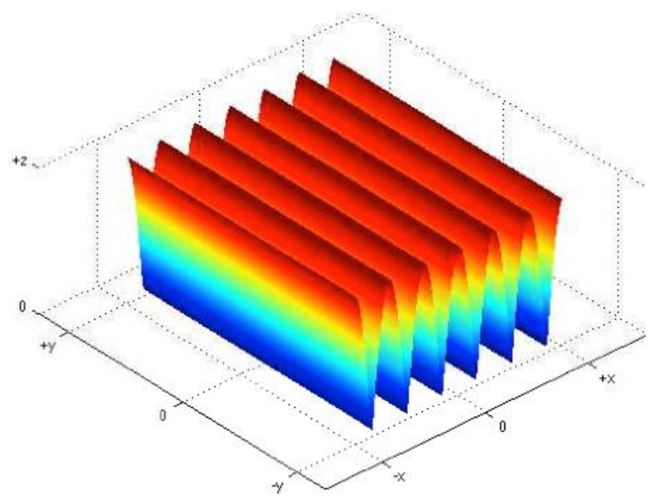


Figure 5.4. The graph for $x[n_1,n_2] = \sin(2\pi n_1/4)$.

5.2.4. Representation of images as 2-D signals

An image can be considered a two-dimensional signal: $g(x, y)$ where x represents the horizontal dimension and y represents the vertical dimension.

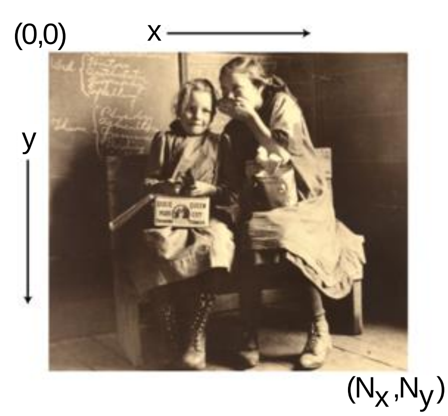


Figure 5.5. Example image and how its represented by a coordinate system.

Continuous-space and continuous-amplitude images can consist of grayscale intensity values, where $g(x, y)$ is a two-dimensional signal representing the grayscale value at location (x, y) where:

$$0 \le x \le N_x \text{ and } 0 \le y \le N_y$$
$$g(x, y) = 0 \text{ represents black}$$
$$g(x, y) = 1 \text{ represents white}$$
$$0 < g(x, y) < 1 \text{ represents proportional gray values}$$

$g(x, y)$ can be displayed as an intensity image or as a mesh graph.

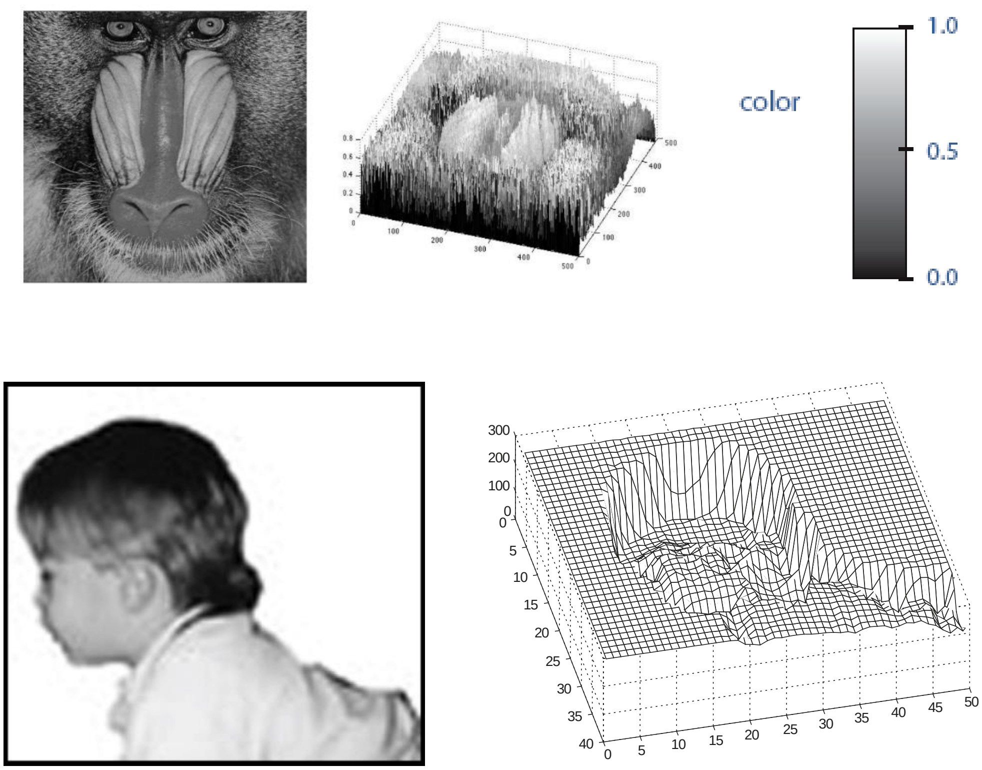


Figure 5.6. Images and their 2-D signal representation.

5.3 The Fourier Transform in 2-Dimensions, spatial frequencies and image signal representation

The Fourier transform $g(x, y)$ has *two* frequency variables: ω_x and ω_y, and is given by:

$$G(\omega_x, \omega_y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x, y) \exp[-j(\omega_x x + \omega_y y)]\, dxdy$$

In this case, the frequency ω_x and ω_y are known as spatial frequencies in the x and y direction. The *magnitude* and *phase* of $G(\omega_x, \omega_y)$ are

$$|G(\omega_x, \omega_y)| \text{ and } \angle G(\omega_x, \omega_y).$$

Let's look at some examples of image representation using Fourier Transform.

5.3.1. Signal representation of images based on Fourier Transform

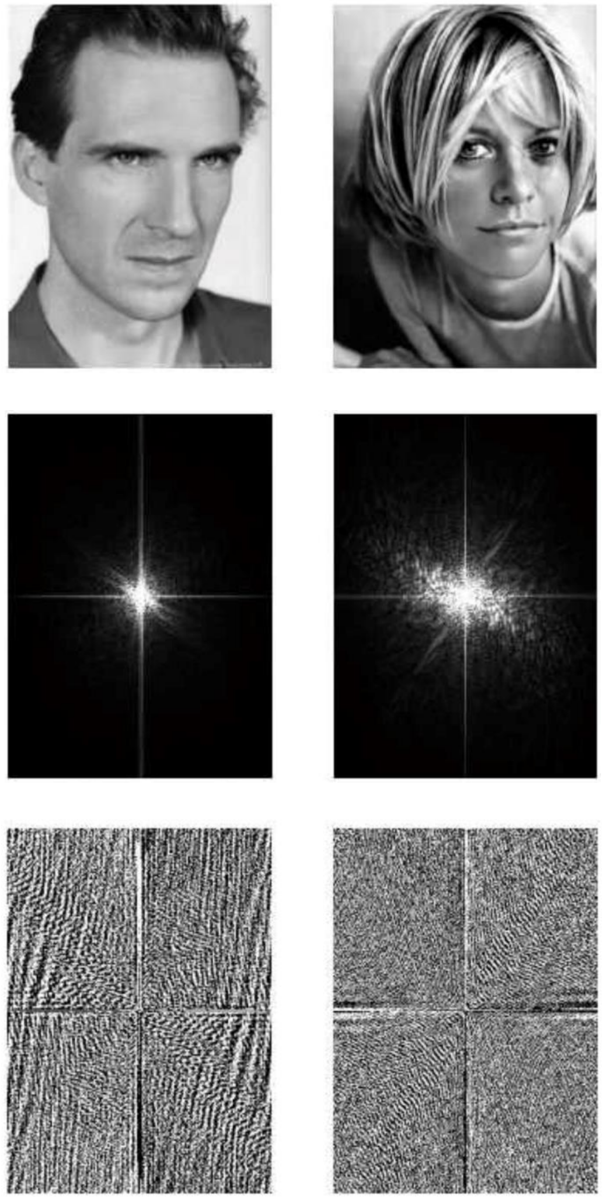


Figure 5.7. Two different intensity images (top left and right) and their respective Fourier transform magnitude $|G(\omega_x, \omega_y)|$ (2nd row) and phase $\angle G(\omega_x, \omega_y)$ (3rd row) images

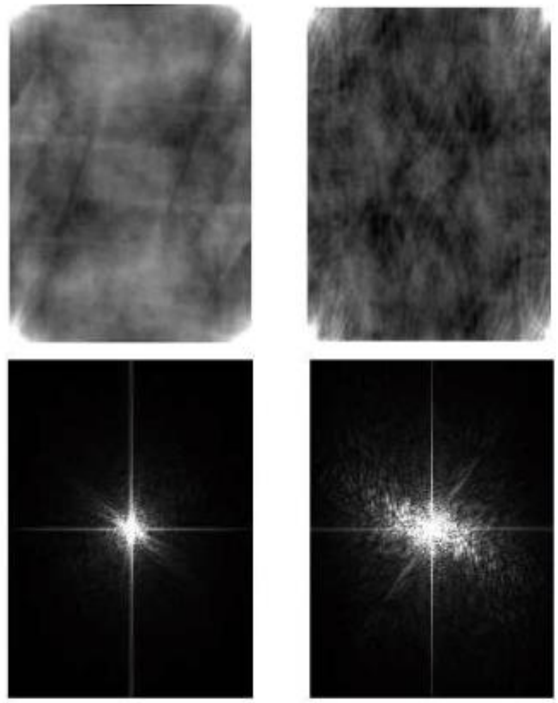


Figure 5.8. Reconstruction using the *magnitude only*: Top left and right images have phase set to be 0

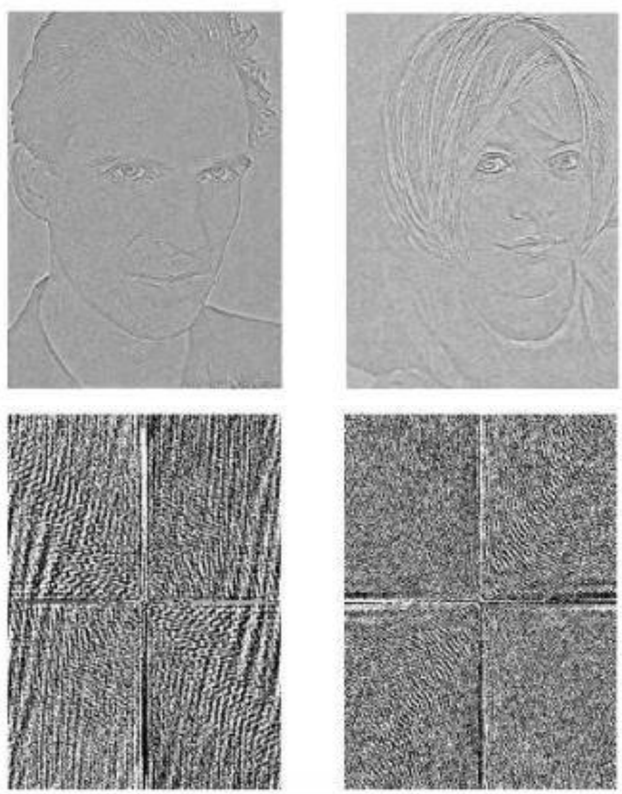


Figure 5.9. Reconstruction using the *phase only*: Top left and right images have a normalized magnitude of 1.

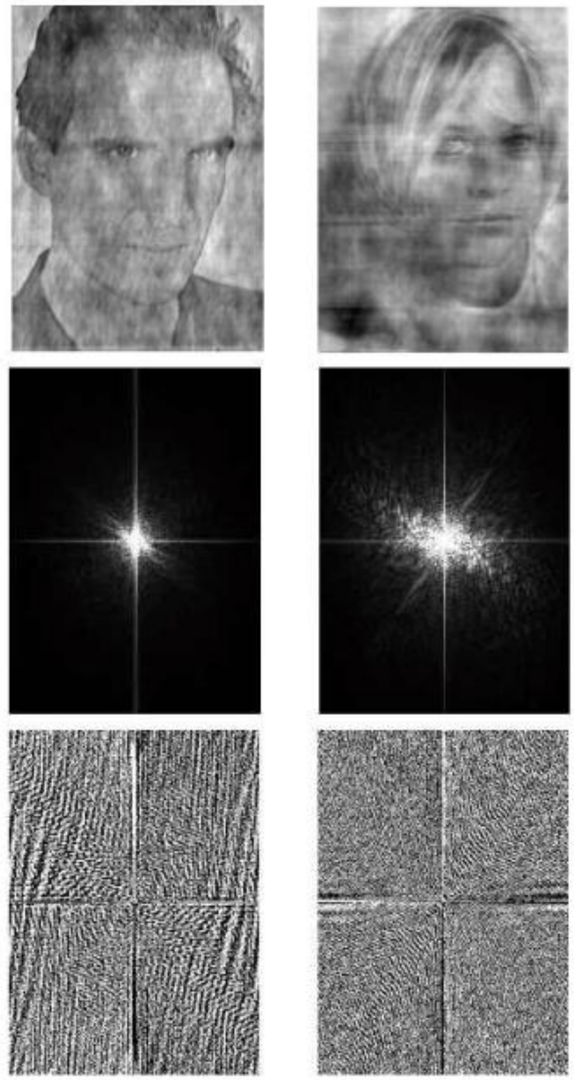


Figure 5.10. Reconstruction swapping magnitude and phase of the images. Top left is the guy's phase and girl's magnitude, and the top right is the girl's phase and the guy's magnitude.

Obviously, the 2-D phase of the image spectrum carries a lot more information than the amplitude spectrum.

5.3.2. 2-D transform of rectangular signals

The 2-D Fourier Transform of a rectangle function

$$g(x,y) = \begin{cases} 1, & \begin{matrix} -a \le x \le a \\ -b \le y \le b \end{matrix} \\ 0, & otherwise \end{cases}$$

is

$$\begin{aligned} G(\omega_x, \omega_y) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y) \exp\left[-j\left(\omega_x x + \omega_y y\right)\right] dxdy \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y) \exp\left[-j\left(\omega_x x + \omega_y y\right)\right] dxdy \\ &= \int_{-b}^{b}\int_{-a}^{a} 1 \cdot \exp\left[-j\left(\omega_x x + \omega_y y\right)\right] dxdy \end{aligned}$$

$$= \int_{-b}^{b} \exp(-j\omega_y y)dy \cdot \int_{-a}^{a} \exp(-j\omega_x x)dx$$
$$= 2b\ sinc\left(\frac{b}{\pi}\omega_y\right) \cdot 2a\ sinc\left(\frac{a}{\pi}\omega_x\right)$$

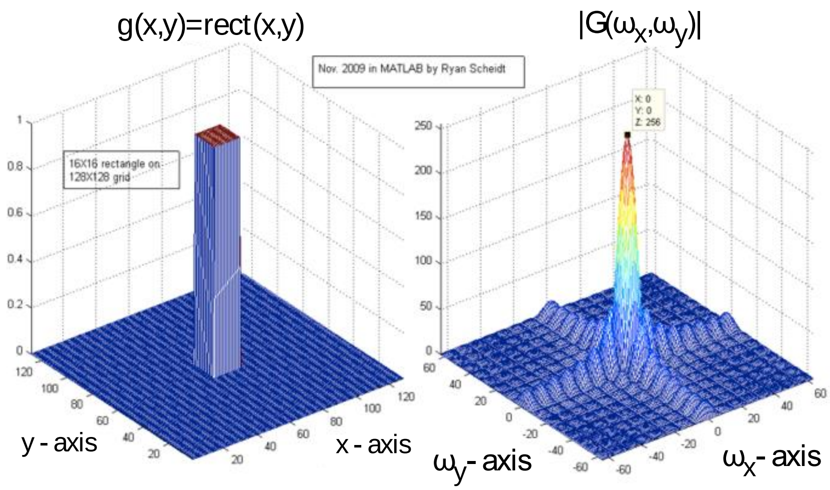


Figure 5.11. 2D rectangular function (left) and its Fourier transform (2 dimensional *sinc* function, right)

5.3.3. 2-D Inverse Fourier Transform

Similar to 1-D signals, the 2-D inverse Fourier Transform is defined as

$$g(x,y) = \frac{1}{(2\pi)^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} G(\omega_x, \omega_y) e^{j(\omega_x x + \omega_y y)} d\omega_x d\omega_y$$

Here are some spectrum and their Inverse Fourier Transform.

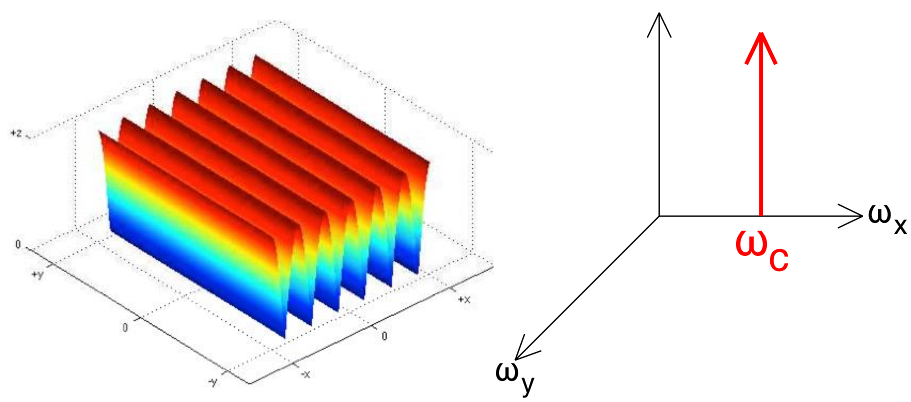


Figure 5.12. X - oscillating sinusoid (left) and its Fourier transform (right), corresponding to the 2-D waveform $g(x,y) = e^{j\omega_c x}$.

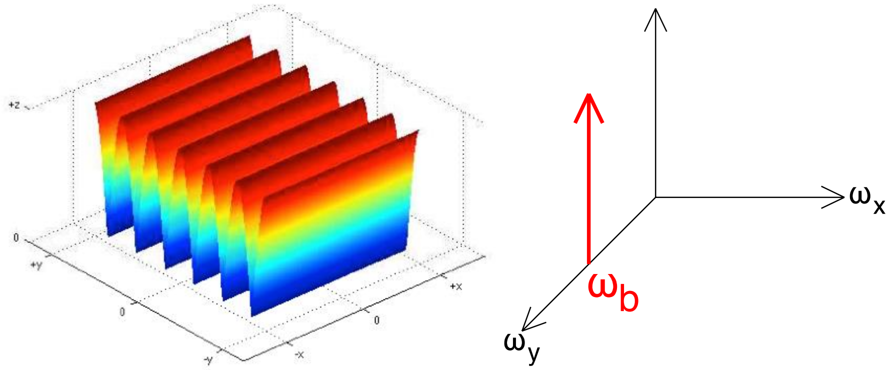


Figure 5.13. Y - oscillating sinusoid (left) and its Fourier transform (right), corresponding to the 2-D waveform $g(x, y) = e^{j\omega_b y}$.

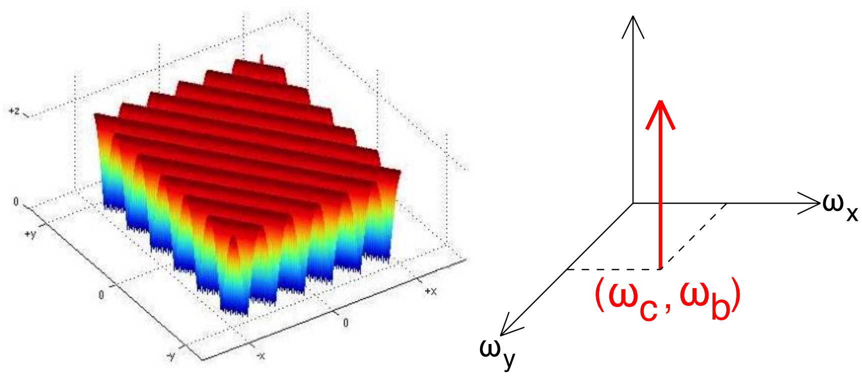


Figure 5.14. Diagonally oscillating sinusoid (left) and its Fourier transform (right), corresponding to the 2-D waveform $g(x, y) = e^{j(\omega_c x + \omega_b y)}$.

Example 5.2 Two-dimensional low-pass filter
If

$$H(\omega_x, \omega_y) = \begin{cases} 1, & \begin{array}{l} -\omega_1 \leq \omega_x \leq \omega_1 \\ -\omega_2 \leq \omega_y \leq \omega_2 \end{array} \\ 0, & otherwise \end{cases}$$

denotes a 2-D low-pass filter with cutoff frequencies ω_1 and ω_2, the inverse Fourier Transform is given by

$$\begin{aligned} h(x, y) &= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} H(x, y) \exp[j(\omega_x x + \omega_y y)]\, dxdy \\ &= \frac{\omega_2 \omega_1}{\pi^2} sinc\left(\frac{\omega_1}{\pi} x\right) sinc\left(\frac{\omega_2}{\pi} y\right) \end{aligned}$$

5.3.4. Efficient Image representation using 2-D Fourier Transform

Uncompressed digital images are typically represented by a 3 matrices of numbers that represent how much red, blue, and green are in the image. For example, an image with a resolution of 500 by 500 pixels will be is represented by a three matrices with 500 by 500. One well known uncompressed image file type is .BMP (bitmap image file). However, bitmaps are known to be extremely large, and as such, it would be desirable to compress these images without compromising image quality too much. One well known file format that compressed bitmaps and other images to smaller sizes is the *JPEG* (Joint Photographic Experts Group) compression algorithm.

First, let's take a look at the 2-D sinusoids with different spatial frequencies in grayscale.

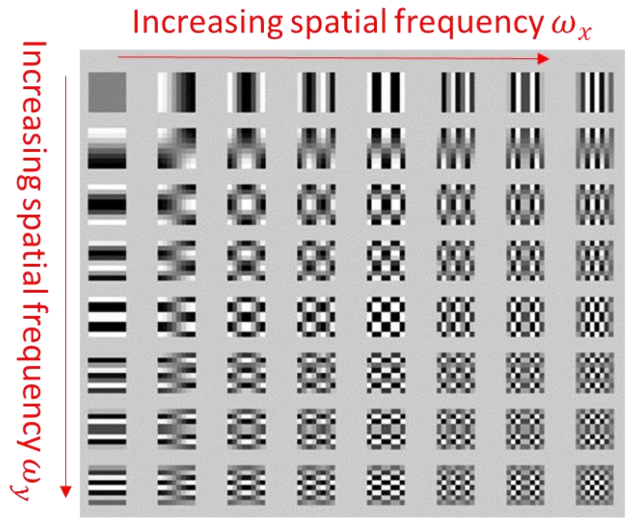


Figure 5.15. 2-D sinusoids $e^{j(\omega_x x+\omega_y y)}$ for different frequencies $\omega_x$and ω_y.

The JPEG algorithm exploits the fact that human eye can only see spatial data below a certain spatial frequency. Consequently, the JPEG algorithm takes an image $g(x,y)$, taking its 2-D Fourier Transform and obtain $G(\omega_x,\omega_y)$, and deleting its highest spatial frequencies from $G(\omega_x,\omega_y)$. The number of frequencies that are deleted depend on how much a user wants to compress the image.

More specifically, the JPEG algorithm takes the three colour space matrices, and breaks them down into 8x8 pixel blocks and then taking the Fourier Transform of those blocks. This results in each 8x8 block of pixels being represented by a weighted version of the 8x8 Fourier Transform coefficients, shown in Figure 5.16. Note how starting from the top left part, the x frequency increases as we proceed to the right, while the y frequency increases as we go from top to bottom. Figure 5.16 below shows us how an example 8x8 image block could be represented by a weighted matrix where each element corresponds to a different frequency.

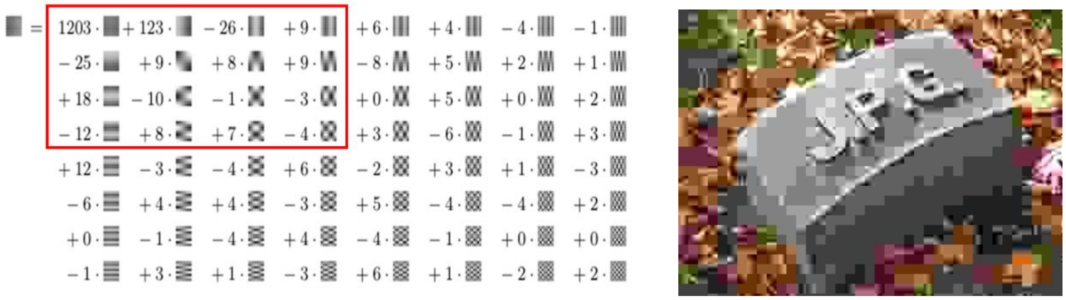


Figure 5.16. 8x8 JPEG images with increasing amount of compression (top to bottom) by storing only the low frequency components of the 2-D signal that represent the image.

The JPEG algorithm compress the image size by only storing the low frequency components in the computer and deleting the high frequency components, thus saving memory space. Since we are not very sensitive to high spatial frequencies, the perceived image quality may not drop significantly. The visual effect to the image will be blurring as shown above. The amount of compression is a trade off between the file size requirement and the required quality of the image. In fact, this whole operation can be thought of us the 2-D image signal go into a system that resembles a *2-D low-pass filter*. The high frequency components disappeared while the low-frequency components remained at the output

Note that image compression is not only beneficial in terms of saving memory space, it is another way of increasing the speed of communications: the time it takes to send a large file is longer than that of a small file.